An Effective Evaluation Tool for Hyperspectral Target Detection: 3D Receiver Operating Characteristic Curve Analysis

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***Abstract*— Receiver operating characteristic (ROC) analysis is performed by a curve, called ROC curve, plotted based on detection probability, P_D, versus false alarm probability, P_F, and has been widely used as an evaluation tool for signal detection. Specifically, the area under an ROC curve (AUC) is calculated and used as a detection measure. Unfortunately, finding distributions of P_D and P_F to generate a continuous ROC curve is practically infeasible. This article investigates approaches to generating a discrete 2D ROC curve of (P_D,P_F) without appealing for probability distributions. Since P_D and P_F are determined by the same threshold τ to specify a detector, an ROC curve of (P_D,P_F) can only be used to evaluate the effectiveness of a detector but not target detectability (TD) and also not background suppressibility (BS). To address this issue, a 3D ROC curve is generated as a function of (P_D,P_F,τ) by introducing a specific threshold parameter τ as a third independent variable. As a result, a 3D ROC curve along with its derived three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) can further be used to design new quantitative measures to evaluate the effectiveness of a detector and its TD and BS. To demonstrate the full utility of 3D ROC analysis in target detection, extensive experiments are performed on two types of targets, prior target detection and anomaly detection, to conduct a comprehensive analysis on 3D ROC curves using new designed detection measures to evaluate target/anomaly detection performance.**

***Index Terms*— 3D receiver operating characteristic (3D ROC), area under an ROC curve (AUC), BKG suppressibility (BS), joint target detectability with BKG suppressibility (JTDBS), overall detection (OD), OD probability (ODP), signal-to-noise probability ratio (SNPR), target detectability (TD), target detection in BKG (TD-BS).**

NOMENCLATURE

3D ROC	Three-dimensional receiver operating characteristic.
AUC $= A_z$	Area under curve.
$A_z^{\text{normalized}}$	Normalized A_z.
AUC_{BS}	$\text{AUC}_{(\text{D,F})} - \text{AUC}_{(\text{D,F})}$.
$\text{AUC}_{(\text{D,F})}$	AUC value of 2D ROC curve of (P_D,P_F).
$\text{AUC}_{(\text{D},\tau)}$	AUC value of 2D ROC curve of (P_D,τ).
$\text{AUC}_{(\text{F},\tau)}$	AUC value of 2D ROC curve of (P_F,τ).
AUC_{MDP}	$\text{AUC}_{\text{SS}} + \text{AUC}_{\text{SF}}$.
AUC_{OD}	$\text{AUC}_{(\text{D,F})} + \text{AUC}_{(\text{D},\tau)} - \text{AUC}_{(\text{F},\tau)}$.
AUC_{ODP}	$\text{AUC}_{(\text{D},\tau)} + [1 - \text{AUC}_{(\text{F},\tau)}]$.
AUC_{SNPR}	$\text{AUC}_{(\text{D},\tau)}/\text{AUC}_{(\text{F},\tau)}$.
AUC_{TD}	$\text{AUC}_{(\text{D,F})} + \text{AUC}_{(\text{D},\tau)}$.
$\text{AUC}_{\text{TD-BS}}$	$\text{AUC}_{(\text{D,F})} - \text{AUC}_{(\text{F},\tau)}$.
BKG	Background.
BS	BKG suppressibility.
CEM	Constrained energy minimization.
CRD	Collaborative RXD.
CRD-DW-STO	l_2-norm minimization and distance Weighted regularization matrix and sum-to-one constraint.
δ^{NP}	Neyman–Pearson detector.
E-CEM	Ensemble-based cascaded CEM.
H-CEM	Hierarchical CEM.
HYDICE	Hyperspectral digital imagery collection experiment.
JTDBS	Joint target detectability with BS.
MDP	Medical diagnosis probability.
OD	Overall detection.
ODP	OD probability.
P_D	Detection probability.
P_F	False alarm probability.
R-AD	Anomaly detector using sample correlation matrix $\mathbf{R}$.
RX-AD	Anomaly detector developed by Reed–Xiaoli.
SNPR	Signal-to-noise probability ratio.
τ	Threshold used by a detector.
TD	Target detectability.
TD-BS	Target detection in BKG.

Manuscript received June 12, 2020; revised August 5, 2020; accepted August 30, 2020. Date of publication September 25, 2020; date of current version May 21, 2021. This work was supported by the Fundamental Research Funds for Central Universities under Grant 3132019341.

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Digital Object Identifier 10.1109/TGRS.2020.3021671

I. INTRODUCTION

RECEIVER operating characteristic (ROC) analysis has been widely used in signal detection theory as an evaluation tool to assess the effectiveness of a given detector [1].

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It is derived from a Neyman–Pearson detector (NPD) which maximizes detection probability, P_D, subject to a constraint β imposed on the false alarm probability, P_F. Varying the constraint β as a parameter on P_F yields different values of P_D. A curve plotted based on P_D versus P_F is called ROC curve of (P_D,P_F) which describes the interaction between P_D and P_F. A tool, which calculates the area under an ROC curve (AUC) and uses AUC values as a criterion, has been used widely in radar detection, medical diagnosis [2]–[4], ecology [5], machine learning [6], and model accuracy [7].

However, there is a practical obstacle in generating such ROC curves, which require prior knowledge of probability distributions to calculate P_D and P_F. In many real-world applications, acquiring probability distributions is nearly impossible and practically infeasible. So without knowing true probability distributions, calculating P_D and P_F poses a great challenge because finding P_D and P_F requires a specific threshold τ that determines a detector to perform detection. Interestingly, this issue has been overlooked in hyperspectral target/anomaly detection. A common practice to generate ROC curves is to find points (P_D,P_F) on the curves by simply varying the threshold τ that determines P_D and P_F. It then uses linear interpolation (including intrapolation and extrapolation) between the found (P_D,P_F) points to generate complete ROC curves. Technically speaking, this practice is incorrect because an ROC curve is nonlinear and convex downward (generally referred to as convex-cap curve). Accordingly, using linear interpolation to approximate an ROC curve is generally not accurate unless there are sufficiently a large number of data samples to make such a nonlinear approximation look like a linear approximation. Unfortunately, this approach has been used extensively in hyperspectral target/anomaly detection where numerous examples of the ROC curve plots have shown zig-zag step curves, many of which are not even convex-cap curves. This article investigates and provides solutions to this issue.

According to the ROC analysis, when $P_F = 0$, P_D must be 0. On the other hand, when $P_F = 1$, P_D must also be equal to 1. Unfortunately, in real cases, this is not true due to the use of a finite number of data samples. To deal with this issue, a normalization process is needed to correct such offset. In terms of normalization, two practical methods are further developed to generate a discrete ROC curve of (P_D,P_F) for a limited number of data samples. One is to use a step size as a threshold τ to calculate (P_D,P_F). The other is to use the true values of data samples as threshold values, τ, to calculate (P_D,P_F). It turns out that most ROC curves reported in the literature were actually generated by step size without normalization. As a consequence, such generated ROC curves result in zig-zag step curves. In fact, a correct ROC curve should be the one using the true data sample values as threshold values τ followed by normalization.

There are also several cautions of using such ROC curves as an analysis tool. A critical one is that both P_D and P_F are calculated by the same threshold τ and they are indeed not independent. As a matter of fact, P_D and P_F interact with each other and they are a functions of an implicit independent parameter specified by the threshold τ. A second one is that the AUC of (P_D,P_F) does not necessarily indicate target detectability (TD) of a detector or false alarm capability. In fact, in many cases the AUC values of (P_D,P_F) can be very high for a detector, but its detection probability, P_D, can be very low and its false alarm probability, P_F, can also be very high as well. This is because both P_D and P_F are determined by the same threshold τ and one affects the other. So, when P_F changes, P_D also changes. A third one is that on some occasions two detectors may yield different ROC curves, but they indeed produce the same AUC value of (P_D,P_F). Under this circumstance, both the detectors have shown the same effectiveness of their detection. Obviously, if both P_D and P_F are calculated independently and used to evaluate their detection performance, these two detectors will perform quite differently. This fact provides evidence that the AUC value of (P_D,P_F) can only be used to evaluate the effectiveness of a detector but not its P_D or P_F. Finally, the last one is its lack of ability in addressing the background (BKG) issue. Since the main focus of signal detection is to detect a signal embedded in noise or BKG, the formulation of NPD is to maximize the detection probability by fixing P_F at a certain level via imposing a constraint β. In this case, varying the constraint β from 0 to 1 yields a complete ROC curve. So, the main concern of NPD is P_D not P_F Unfortunately, it is P_F that determines BKG suppression.

To deal with the above-mentioned issues arising in 2D ROC analysis, a 3D ROC analysis was first envisioned and applied to hyperspectral subpixel detection in [9] where a 3D ROC curve was introduced as a function of three parameters, P_D,P_F, and a threshold parameter τ. Its idea includes τ as an independent variable to produce P_D,P_F as its dependent functions and then generates an ROC curve in a 3D space of (P_D,P_F,τ). The resulting ROC curve is called the 3D ROC curve. By virtue of such a 3D ROC curve, three 2D ROC curves can also be generated, called D ROC curve of (P_D,P_F), 2D ROC curve of (P_D,τ), and 2D ROC curve of (P_F,τ) by projecting a 3D ROC curve on their own respective planes, respectively. By taking advantage of the AUC values calculated from these three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ), referred to as $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$, respectively, these AUC values can be further used to evaluate 1) the effectiveness of a detector; 2) detection probability of a detector; and 3) BKG suppression (BS) of a detector, respectively.

Since its inception in [8], the 3D ROC analysis has been used in medical imaging to evaluate the effectiveness of medical diagnosis such as magnetic resonance imaging [9], [10] and breast imaging [11]. Furthermore, by taking advantage of its ability in detection of abundance fractions, the 3D ROC analysis was even applied to biological agent detection [12] and water monitoring [13]. Moreover, by interpreting the threshold parameter τ as a system configuration parameter, 3D ROC analysis was further applied to biometric recognition analysis [14], [15]. In addition, the 3D ROC analysis has also found its applications in vital sign signal analysis in medical data [16], [17]. A detailed survey and review on 3D ROC analysis can be found in [18] with detailed discussions in [19, Ch. 3]. Most recently, the 3D ROC analysis was used to evaluate hyperspectral anomaly

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detection performance [20], [21] and has been further used as an evaluation tool for hyperspectral image classification [22].

However, all the above-mentioned works considered the three 2D ROC curves derived from a 3D ROC curve as separate curves for individual data analysis and did not fully take advantage of the key idea behind the 3D ROC analysis such as inter-relationship among the three 2D ROC curves. This article revisits the 3D ROC analysis and further explores the full utility of the 3D ROC analysis in hyperspectral target/anomaly detection. In particular, several new 3D ROC curve-driven quantitative detection measures are further derived by different combinations of the AUC values of the three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) to evaluate a detector's effectiveness, TD, BS, specifically, TD-BS, signal-to-noise probability ratio (SNPR), overall detection probability (ODP), and overall detection (OD), all of which can effectively evaluate hyperspectral target/anomaly detection quantitatively.

As a final conclusion, we summarize major contributions of this article as follows.

1) One of main contributions is to explore insights into generating discrete ROC curves and to develop two practical methods generating correct ROC curves with normalization. On many occasions, the AUC values of (P_D,P_F) of two detectors are very close with negligible differences without normalization. However, their difference can be more appreciable if their ROC curves of (P_D,P_F) are correctly generated with normalization.
2) Another important contribution is to analyze the role of each of the three 2D ROC curves, 2D ROC curve of (P_D,P_F), 2D ROC curve of (P_D,τ), and 2D ROC curve of (P_F,τ) generated from a 3D ROC curve in target/anomaly detection. Such study has never been investigated in the literature and even not exploited in [8]–[22]. Particularly, an ROC curve of (P_D,P_F) can only be used to evaluate "the effectiveness of a detector in detection performance" but not for "its TD," P_D or "its BS," P_F. This is because the same threshold τ determines both P_D and P_F, in which case two detectors may have the same AUC value of (P_D,P_F) with different AUC values of (P_D,τ) and (P_F,τ). To resolve this issue, the 2D ROC curve of (P_D,τ) can be used to dictate detectability of a detector as τ varies across the range. Conversely, whether a detector is effective is not solely determined by P_D but also by P_F which is mainly caused by either noise or BKG. The 2D ROC curve of (P_F,τ) happens to be the one able to measure the degree of noise/BKG being suppressed. The reason that the 2D ROC curve of (P_D,P_F) cannot be used for BKG suppression is the same reason that it cannot be used to measure detectability of a detector because both P_D and P_F use the same threshold τ.
3) A novel and unique contribution derived from this article is design and development of four different types of new detection measures: 1) TD to evaluate effectiveness of a detector in target detection; 2) BS to assess capability of a detector in BKG suppression; 3) JTDBS to consider joint ability of a detector in detection of targets/anomalies embedded in BKG; and 4) SNPR to measure the ratio of signal detection probability, $\mathrm{AUC}_{(D,\tau)}$, to false alarm probability, $\mathrm{AUC}_{(F,\tau)}$, a similar idea to signal-to-noise ratio (SNR) commonly used in communications/signal processing. It turns out that SNPR is the most effective detection criterion among all the developed detection measures.

II. Binary Hypothesis Testing

A. Neyman–Pearson Detection Theory

A signal detection problem is generally described by a binary hypothesis testing problem as follows [1]:

$$\begin{aligned} &H_0 : Y \approx p_0(y) \\ &\quad \text{versus} \\ &H_1 : Y \approx p_1(y) \end{aligned} \tag{1}$$

where $p_0(y)$ and $p_1(y)$ are the probability density functions (pdfs) that govern the two hypotheses H_0 and H_1, called "null hypothesis" which represents the signal absence and "alternative hypothesis" which represents the signal presence, respectively. An NPD, denoted by $\delta^{\mathrm{NP}}(\mathbf{r})$, is designed to solve the following constrained optimization problem:

$$\max_{\delta}\{P_D(\delta)\} \text{ s.t. } P_F(\delta) \le \beta \text{ for any given } 0 \le \beta \le 1 \tag{2}$$

where β is prescribed as the significant level of the detector. The maximum in (2) is found over all possible detectors, $\delta(\mathbf{r})$. The well-known Neyman–Pearson lemma shows that the optimum solution to (2) is a likelihood ratio test, $\Lambda(\mathbf{r})$, given by

$$\delta^{\mathrm{NP}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \Lambda(\mathbf{r}) > \tau \\ 1 & \text{with probability } \gamma \text{ if } \Lambda(\mathbf{r}) = \tau \\ 0; & \text{if } \Lambda(\mathbf{r}) < \tau \end{cases} \tag{3}$$

where $\Lambda(\mathbf{r}) = p_1(\mathbf{r})/p_0(\mathbf{r})$, the threshold τ is determined by the constraint β, and γ is the probability of saying H_1 given by

$$\gamma = \frac{\beta - P_0(\{y \in \Gamma | p_1(y) > \tau p_0(y)\})}{P_0(\{y \in \Gamma | p_1(y) = \tau p_0(y)\})} \tag{4}$$

when $\Lambda(\mathbf{r}) = \tau$. Using (3), P_D and P_F can be found by

$$P_D = \int_{\Lambda(\mathbf{r})>\tau} p_1(\mathbf{r}) d\mathbf{r} + \gamma\, P(\{\mathbf{r} | \Lambda(\mathbf{r}) = \tau\}) \tag{5}$$

$$P_F = \int_{\Lambda(\mathbf{r})>\tau} p_0(\mathbf{r}) d\mathbf{r} + \gamma\, P(\{\mathbf{r} | \Lambda(\mathbf{r}) = \tau\}) \tag{6}$$

with τ determined by $P_F = \beta$ via (6).

As an illustrative example, Fig. 1 plots the four decisions made by NPD where $P_{TN} = 1 - P_F$, referred to as true negative (TN) corresponding to signal absence and $P_M = 1 - P_D$ referred to as miss probability corresponding to no signal detected when there is indeed a signal present. Details of Neyman–Pearson detection theory can be found in [1].

From Fig. 1, we can construct the following matrix in Table I which corresponds to a confusion matrix commonly used in binary classification.

It is worth noting that in many applications such as medical diagnosis, P_{TN} in Table I can also be used as a measure,

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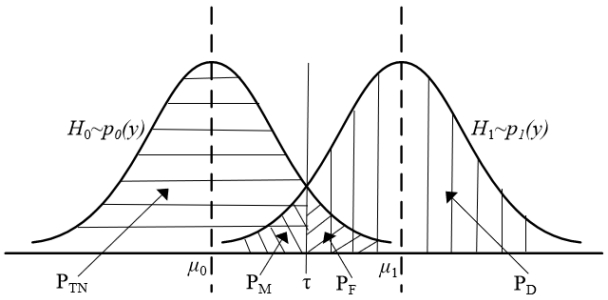


Fig. 1. Illustration of probabilities P_D, P_F, P_M, and P_{TN}.

TABLE I
CONFUSION MATRIX FOR BINARY CLASSIFICATION

classification \ true	noise	signal
noise	P_{TN}	P_M
signal	P_F	P_D

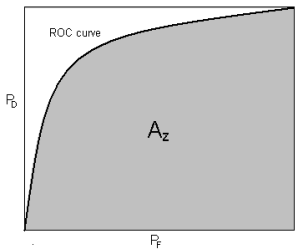


Fig. 2. Example of AUC, A_z, calculated by an ROC curve.

referred to as the specificity (SF) as opposed to detection probability, P_D, which is referred to as the sensitivity or true positive (TP), while P_F and P_M are referred to as the false negative (FN) and false positive (FP), respectively.

B. ROC Analysis

According to (2), the detection performance of an NPD in (3) is determined by P_D subject to the constraint on P_F, which is bounded from above by a specific given value β. To be precise, for a given β we can calculate P_F using (6) to find the threshold τ. Then, this same τ is further used to calculate P_D via (5). So a different value of β also produces different values of P_F and P_D. That is, for each value of β, it generates a pair of (P_D, P_F) as a point on an ROC curve where a threshold τ can be found by (7) by setting $P_F = \beta$ and then both P_D and P_F are calculated by the same τ produced by (5) and (6). If we now vary the value of β from 0 to 1, then we can plot a curve of P_D versus P_F as shown in Fig. 2, to be called the ROC curve, which is a convex downward curve, referred to as a convex-cap curve.

This ROC curve allows users to assess the effectiveness of a detector as a function of P_D versus P_F through an implicit parameter, threshold τ. In many applications, it is necessary to compare one detector against another in terms of their detection performance. Visually inspecting their ROC curves can only provide a qualitative analysis. To deal with this issue, an alternative approach to the use of ROC curves is to calculate the area under an ROC curve, called AUC, A_z which is particularly useful in medical diagnosis [2]–[4]. This AUC can be used to evaluate the effectiveness of a detector quantitatively. As a result, the higher the AUC of a 2D ROC curve of (P_D, P_F), the better the detector.

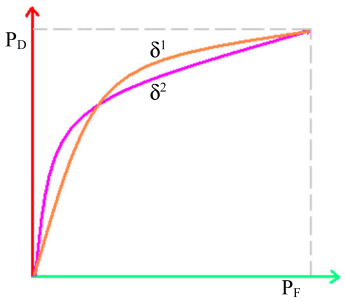


Fig. 3. Example of ROC curves generated by two detectors with the same A_z.

However, there is a dilemma arising from AUC. On some occasions, two detectors δ^1 and δ^2 may generate two different 2D ROC curves but have the same AUC, denoted by A_z, as shown in Fig. 3.

In this case, both δ^1 and δ^2 may yield the same detection performance even when they have different ROC curves. In other words, using AUC as a criterion may not be a good measure to evaluate the two detectors δ^1 and δ^2 in Fig. 3 since they have the same value of AUC but indeed have different values of detection probability P_D for any given same false alarm probability, P_F. The 3D ROC analysis to be presented in Section IV is particularly developed to address this issue.

As a final comment, we would like to make an important point. For continuous pdfs, $p_0(y)$ and $p_1(y)$ in (1), $p_0(\{y \in \Gamma | p_1(y) = \tau p_0(y)\})$ is generally 0, and thus, γ in (4) will be 0 as well. However, for discrete probability mass functions, $p_0(y)$ and $p_1(y)$, $p_0(\{y \in \Gamma | p_1(y) = \tau p_0(y)\})$ is usually greater than 0, and also implies that γ in (4) will also be greater than 0. This γ is exactly the cause that a discrete 2D ROC curve turns out to be a zig-zag step curve. More specifically, there may have two or more points of (P_D, P_F) which have the same value of P_F but different values of P_D corresponding to different points lined up vertically as a jump or corresponding to different points lined up horizontally as a step line segment. As a consequence of $\gamma > 0$, the generated discrete 2D ROC curve will not be a convex-cap curve like Fig. 2 but rather than look like a step-size climbing curve, which has been seen in most research works for target detection reported in the literature. Unfortunately, this fact has been taken for granted and very little work has been devoted to investigation on this phenomenon.

III. HOW TO GENERATE ROC CURVES

A. Issues in Generating Discrete 2D ROC Curves

As noted above, in many real-world applications finding probability distributions based on a limited number of data samples for a hypothesis testing problem may not be realistic. Accordingly, using probability distributions to generate an ROC curve of (P_D, P_F) is not practical. So a common approach is to use the threshold parameter τ as a variable to calculate

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both P_D and P_F. Since each τ can only produce a pair of P_D and P_F as a point of the ROC curve of (P_D,P_F), we need to vary τ to generate a complete ROC curve of (P_D,P_F). Unfortunately, this is easily said than done. The main issue arises in how to model the threshold τ. Despite that there are various sources of software available in the public domain such as MATLAB that can be used for this purpose, the detailed information of its implementation is limited. So there is no guarantee that the produced ROC curves are really what these curves should be. In particular, users generally generate ROC curves using a step size. Its consequence is that the generated ROC curves look like zig-zag step-size curves as shown in many research publications due to the use of linear interpolation, which is incorrect since an ROC curve is nonlinear and convex-cap.

Because there are no probability distributions, $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$, in (4) and (5) available in practical applications, directly using the likelihood ratio test in (3) is not feasible. In this case, we rewrite (3) as

$$\delta_\tau^{\mathrm{NP}}(\mathbf{r}) = \begin{cases} 1; & \text{if } \delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) \geq \tau \\ 0; & \text{if } \delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) < \tau \end{cases} \tag{7}$$

where

$$\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}) = \frac{\delta_\tau^{\mathrm{NP}}(\mathbf{r}) - \min_{\mathbf{r}} \delta_\tau^{\mathrm{NP}}(\mathbf{r})}{\max_{\mathbf{r}} \delta_\tau^{\mathrm{NP}}(\mathbf{r}) - \min_{\mathbf{r}} \delta_\tau^{\mathrm{NP}}(\mathbf{r})} \tag{8}$$

and τ is determined by step size or data sample values which in turn determines $P_F = \beta$. Due to normalization, $\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r})$ in (8) takes values in the range of [0, 1] with $0 \leq \delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) \leq 1$. We then can use $0 \leq \tau \leq 1$ in (7) as the threshold to calculate P_D and P_F instead of directly calculating $P_F = \beta$ to find P_D generating a discrete 2D ROC curve of (P_D,P_F).

Or alternatively, we can use another detector

$$\delta_\tau^{\mathrm{NP}}(\mathbf{r}) = \begin{cases} 1; & \text{if} \delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) > \tau \\ 0; & \text{if} \delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) \leq \tau \end{cases} \tag{9}$$

to replace (7) by making $\delta_\tau^{\mathrm{NP}}(\mathbf{r}) = 0$ instead of $\delta_\tau^{\mathrm{NP}}(\mathbf{r}) = 1$ in (7) when $\delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) = \tau$.

As noted in (4), when $\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}) = \tau$, the detector $\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}) = \tau$ can go either way by making decision 1 to detect the presence of a signal as (7) does or 0 to declare the absence of a signal as (9) does. To satisfy the constraint, $P_F = \beta$, γ is generally not 0 but greater than 0. So, technically speaking, an optimal detector should be a randomized detector, $\delta_{\mathrm{normalized}}^{\mathrm{RNP}}(\mathbf{r})$, specified by

$$\delta_{\mathrm{normalized}}^{\mathrm{RNP}}(\mathbf{r}) = \gamma\, \delta_{\mathrm{normalized}}^{\mathrm{NP}(1)}(\mathbf{r}) + (1-\gamma)\delta_{\mathrm{normalized}}^{\mathrm{NP}(0)}(\mathbf{r}) \tag{10}$$

where $\delta_{\mathrm{normalized}}^{\mathrm{NP}(1)}(\mathbf{r})$ and $\delta_{\mathrm{normalized}}^{\mathrm{NP}(0)}(\mathbf{r})$ are the detectors specified by (7) and (9), respectively, and $0 < \gamma < 1$ is the probability of choosing the detector $\delta_{\mathrm{normalized}}^{\mathrm{NP}(1)}(\mathbf{r})$ and $0 < 1-\gamma < 1$ is the probability of choosing the detector $\delta_{\mathrm{normalized}}^{\mathrm{NP}(0)}(\mathbf{r})$.

So for continuous pdfs, the probability $\gamma = 0$ and no such issue arises. Thus, both the detectors, $\delta_{\mathrm{normalized}}^{\mathrm{NP}(1)}(\mathbf{r})$ and $\delta_{\mathrm{normalized}}^{\mathrm{NP}(0)}(\mathbf{r})$ produce the same identical 2D ROC curve. However, this is no longer true for discrete pmfs with $0 < \gamma < 1$. Both $\delta_{\mathrm{normalized}}^{\mathrm{NP}(1)}(\mathbf{r})$ and $\delta_{\mathrm{normalized}}^{\mathrm{NP}(0)}(\mathbf{r})$ generate two different 2D ROC curves as will be shown in the experiments conducted in Section VI (see Figs. 8 and 13). Detailed discussions on randomized detectors in (10) can be found in minimax detection and NPD theories [1].

B. Using Data Sample Values as Threshold Values of τ

There are two ways to generate a discrete 2D ROC curve of (P_D,P_F). One is to use a step size to divide [0, 1] for τ. The other is to use the values of normalized detectors obtained from real data samples which will be used to serve the value of τ.

It should be noted that for each data sample, its detected normalized value can be used as the threshold τ. If there are a total number of N data samples, then there will be N values of τ to be used to generate ROC curves. The resulting generated ROC curves will be more accurate than that using a step size for τ to generate ROC curves. It is often case that such found τ may not correspond to the true value of a data sample. So the case of $\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}) = \tau$ in (7) and (9) may not happen. Since both $0 \leq \tau \leq 1$ and $0 \leq \delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) \leq 1$ are real-valued, τ can be the values of the normalized detector obtained from data samples. To take care of this real situation, we need to use τ to run through the values of all data samples to include all possible cases of $\delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}) = \tau$ to generate ROC curves. Generally, it is the case that $\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}) = \tau$ makes significant difference. In other words, there is a significant difference in generating ROC curves on the decision made on $\delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}) = \tau = 0$ or 1 between using (7) and (9).

Let $P_D(\tau)$ and $P_F(\tau)$ be P_D and P_F using the threshold τ, respectively. Also, let n_{target} be the total number of target pixels and $\{\mathbf{r}_i^{\mathrm{target}}\}_{i=1}^{n_{\mathrm{target}}}$ be arranged in accordance with $|\delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}_i^{\mathrm{target}})| \leq |\delta_{\mathrm{normalized}}^{\mathrm{NP}}(\mathbf{r}_{i+1}^{\mathrm{target}})|$ for $1 \leq i \leq n_{\mathrm{target}}$. Also, let $\mathbf{r}_i^{\text{non-target}}$ be a nontarget data sample pixel not in $\{\mathbf{r}_i^{\mathrm{target}}\}_{i=1}^{n_{\mathrm{target}}}$. According to P_D,P_M, P_F, and P_{TN}, we can also define their corresponding number of decisions.

1) $n_{D,\tau} = \left| \left\{ \delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}_i^{\mathrm{target}}) = 1 \right\}_{\mathbf{r}_i^{\mathrm{target}} \in \{\mathbf{r}^i\}_{i=1}^N} \right|$.
2) $n_{M,\tau} = \left| \left\{ \delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}_i^{\mathrm{target}}) = 0 \right\}_{\mathbf{r}_i^{\mathrm{target}} \in \{\mathbf{r}^i\}_{i=1}^N} \right|$.
3) $n_{F,\tau} = \left| \left\{ \delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}_i^{\text{non-target}}) = 1 \right\}_{\mathbf{r}_i^{\text{non-target}} \in \{\mathbf{r}^i\}_{i=1}^N} \right|$.
4) $n_{TN,\tau} = \left| \left\{ \delta_{\mathrm{normalized},\tau}^{\mathrm{NP}}(\mathbf{r}_i^{\text{non-target}}) = 0 \right\}_{\mathbf{r}_i^{\text{non-target}} \in \{\mathbf{r}^i\}_{i=1}^N} \right|$.

When either (7) or (9) is used, their detection probability, $P_D(\tau)$, and false alarm probability, $P_F(\tau)$, can be calculated as follows:

$$P_D(\tau) = a_\tau = \frac{n_{D,\tau}}{n_{D,\tau} + n_{M,\tau}} = \begin{cases} 1 & \text{if } n_{M,\tau} = 0 \\ < 1 & \text{if } n_{M,\tau} \neq 0 \end{cases} \tag{11}$$

$$P_F(\tau) = b_\tau = \frac{n_{F,\tau}}{n_{F,\tau} + n_{TN,\tau}} = \begin{cases} 0 & \text{if } n_{F,\tau} = 0 \\ > 0 & \text{if } n_{F,\tau} \neq 0. \end{cases} \tag{12}$$

As a result of (11) and (12), the original 2D ROC curve of (P_D,P_F) starts the point $(a_{\tau=1}, b_{\tau=1})$ and end up with $(a_{\tau=0}, b_{\tau=0})$ and its AUC, A_z, should be calculated by the region bounded by the box specified by its length $(a_{\tau=0} - a_{\tau=1})$ at the horizontal axis and its height $(b_{\tau=0} - b_{\tau=1})$ at the vertical

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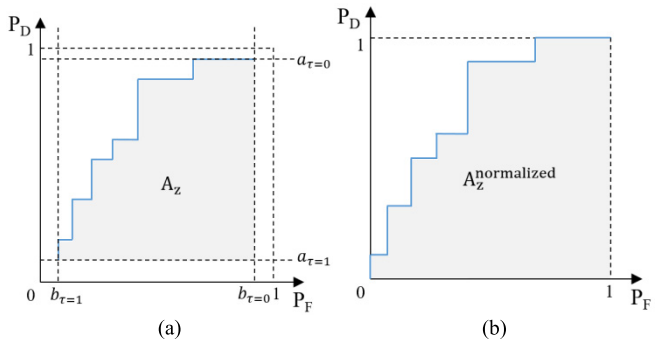


Fig. 4. Original discrete ROC curve and its normalized ROC curve. (a) A_z. (b) Normalized A_z by (13).

axis. Accordingly, its normalized $A_z^{\text{normalized}}$ can further be calculated by

$$A_z^{\text{normalized}} = A_z/(a_{\tau=0} - a_{\tau=1})(b_{\tau=0} - b_{\tau=1}) \tag{13}$$

which is the corrected AUC of the 2D ROC curve of $(P_{\text{D}},P_{\text{F}})$. This is because the normalization carried out by (12) is not by linear interpolation but rather convex-interpolation as it should be. Fig. 4(a) and (b) shows the AUC, A_z, of the original 2D ROC curve and its normalized A_z, respectively. As shown in Fig. 4, a discrete 2D ROC curve is bounded by $(a_{\tau=1}, b_{\tau=1}, a_{\tau=0}, b_{\tau=0})$ with a box specified by $(a_{\tau=1}, b_{\tau=1})$, $(b_{\tau=1}, a_{\tau=0})$, $(b_{\tau=0}, a_{\tau=1})$, and $(a_{\tau=0}, b_{\tau=0})$.

It should be noted that a crucial difference between using (3) and (7) is the randomized rule specified by the probability γ in (4). For continuous pdfs used in (5) and (6), $\gamma = 0$, that is, $n_{\text{D},\tau} = 0$, $n_{\text{M},\tau} = 0, n_{\text{F},\tau} = 0$, and $n_{\text{TN},\tau} = 0$. But for discrete probability mass functions, γ is generally greater than 0. It is this γ that causes a discrete 2D ROC curve $(P_{\text{D}},P_{\text{F}})$ to look like Fig. 4(a) with $a_{\tau=1} > 0$, $a_{\tau=0} < 1$, $b_{\tau=1} > 0$, and $b_{\tau=0} < 1$.

Finally, coincidentally, MATLAB, indeed, uses this data sample-based thresholding approach to produce a discrete 2D ROC curve. However, it does also have the same issue that $a_{\tau=1} > 0$, $a_{\tau=0} < 1$, $b_{\tau=1} > 0$, and $b_{\tau=0} < 1$. To deal with this problem, it uses linear extrapolation to connect (0, 0) to ($a_{\tau=1} > 0$ and $b_{\tau=1} > 0$) and ($a_{\tau=0} < 1$ and $b_{\tau=0} < 1$) to (1, 1) to make a complete ROC curve from (0, 0) and (1, 1). So technically, it is found that AUC is not really correct since the ROC curve must be normalized.

C. Using Step Size as Threshold Intervals for τ

A common and general approach to generating ROC curves is to use a step size Δ to divide $0 \le \tau \le 1$ into small intervals with interval length Δ and use one of the end points of each interval as the value of τ to calculate $P_{\text{D}}(\tau)$ and $P_{\text{F}}(\tau)$ which are P_{D} and P_{F} resulting from using the threshold τ, respectively.

So let $\{\mathbf{\Delta}_k\}_{k=1}^K$ be a set of variable step sizes and assume that $\{\tau_k\}_{k=1}^K$ are K thresholds with $\tau_k = \tau_{k-1} + \mathbf{\Delta}_k$ for $1 \le k \le K$, $\tau_0 = 0$, and $\tau_K = 1$. Then, there are K subintervals, $\{[\tau_{k-1}, \tau_k)\}_{k=1}^K$ with $K+1$ end points specified by $\{\tau_k\}_{k=0}^K$. Now, if we assume that the magnitudes of $\{\delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_i)\}_{i=1}^N$ are arranged in strictly ascending order without loss of generality, that is, $\delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_k) < \delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_{k+1})$ for $1 \le k \le K$. In this case, we can set $\tau_k = \delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_k)$ and $\mathbf{\Delta}_k = \delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_{k+1}) - \delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_k) = \tau_{k+1} - \tau_k$. Then, this is reduced to the case of using the data sample values as threshold values discussed above.

For simplicity, a uniform step size Δ is generally used. For example, if $\Delta = 0.01$, then there are only 101 points to be used to generate ROC curves. However, if the image size is $N = 200 \times 200 = 4 \times 10^4$ and using the data sample values as threshold values, then there are 40 000 data samples that can be used as τ values to generate ROC curves. Comparing to the step-size-generated ROC curves which are generated by each step-size-determined τ not necessarily corresponding to a true value of data sample, such real data value-generated ROC curves can reflect the true values of ROC curves. To avoid the above scenarios, we can choose $\Delta_{\min} = \min_{1\le i\le N-1}(\delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_i) - \delta^{\text{NP}}_{\text{normalized}}(\mathbf{r}_{i+1}))$. This case will produce more $(P_{\text{D}},P_{\text{F}})$ points than the size of data samples. As a result, the generated ROC curve will look like Fig. 2.

Finally, the detailed step-by-step procedure of generating a 2D ROC curve is given below.

Practical Procedure for Generating a 2D ROC Curve of $(P_{\text{D}},P_{\text{F}})$

1. First normalize the detector $\delta^{\text{NP}}_{\tau}(\mathbf{r})$ in (7) to $\delta^{\text{NP}}_{\text{normalized},\tau}(\mathbf{r})$ in (7).
2. For each $0 \le \tau \le 1$, we use $\delta^{\text{NP}}_{\text{normalized},\tau}(\mathbf{r})$ in (7) to calculate P_{D} and P_{F}. It should be noted that the x-axis of P_{F} is calculated according to the reverse order of τ, that is, P_{F} is a function of $1 - \tau$.
3. For each pair of the calculated $(P_{\text{D}},P_{\text{F}})$ in step 1 for one value of τ, a point of 2D ROC curve of $(P_{\text{D}},P_{\text{F}})$ can be generated.
4. By varying τ from 0 to 1, pairs of τ-specified $(P_{\text{D}},P_{\text{F}})$ can be calculated which can be used to generate a 2D ROC curve of $(P_{\text{D}},P_{\text{F}})$ with P_{F} as x-axis from 0 to 1 and P_{D} as y-axis from 0 to 1.
5. Specifically, use (11) and (12) to find $(a_{\tau=1}, b_{\tau=1})$ and $(a_{\tau=0}, b_{\tau=0})$.
6. Normalize A_z by (13).

To conclude this section, three comments are noteworthy.

1) Since a discrete ROC curve is generated by a finite number of $(P_{\text{D}},P_{\text{F}})$ points, it requires interpolation (intrapolation or extrapolation) to approximate a real-valued ROC curve across the range from 0 to 1. On the other hand, an ROC curve is convex-cap, and such interpolation should be performed nonlinearly. Using linear interpolation to generate an ROC curve is technically inappropriate. As a result, two phenomena generally occur in ROC curve plots reported in the literature. One is that ROC curves are zig-zag step curves and may not also be convex-cap. The other is that a_τ in (11) and b_τ in (12) may not always be 0 or 1. These cases are inevitable when a discrete ROC curve is generated.

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Nevertheless, using data sample values as the threshold values τ can offer relief to some extent. Unfortunately, these two issues have never been investigated in the past.

2) Unlike a continuous 2D ROC curve which is a convex-cap curve calculated by probability distributions, a discrete 2D ROC curve is calculated by (7), (10), and (11) using data sample values or step-size as thresholds to find discrete (P_D,P_F) points which must be interpolated by convex-cap functions. In practice, this is generally done by linear interpolation which results in a step-size curve. However, as the number of data samples goes sufficiently large or the step size goes sufficiently small, the convex interpolation will be approximated more closely by linear interpolation. Accordingly, a discrete-valued ROC curve will converge to a real-valued ROC curve. This also indicates that $(a_{\tau=1}, b_{\tau=1})$ will approach (0, 0) and $(a_{\tau=0}, b_{\tau=0})$ will also be close to (1, 1) as it should be like the one in Fig. 2.
3) Since step size-determined points used to plot a 2D ROC curve of (P_D,P_F) are generally not too many and less than a few hundreds points, such generated 2D ROC curve can be considered as a discrete-valued 2D ROC curve. On the other hand, the number of data samples used to plot a 2D ROC curve of (P_D,P_F) is usually very large and much more than the number of step-size-determined points. Thus, the data sample-generated 2D ROC curve can be considered as a contiguous-valued 2D ROC curve. Consequently, it is more accurate than the step size-determined discrete-valued 2D ROC curve. Nevertheless, both discrete-value and contiguous-valued 2D ROC curves are still discrete and suffer from the issue arising from the use of randomized probability γ in (4) which is generally greater than 0.

IV. 3D ROC Analysis

According to NPD, its performance is determined by the significance of level, β, which is specified by P_F and then β in turn determines P_D. But, in fact, β is actually determined by the threshold τ which corresponds to the unknown cost incurred by a detector. As a consequence, both P_D and P_F are indeed determined by the unknown threshold τ, more specifically, functions of τ. A 3D ROC curve simply a 3D plot of (P_D,P_F,τ) includes τ as an independent variable and P_D and P_F as dependent variables through τ.

From the hypothesis testing problem specified by (1), a detector makes a binary hard decision by thresholding a real-valued LRT, $\Lambda(\mathbf{r})$, via a threshold τ in (3). Accordingly, the detector performance is actually determined by two parameters, $\Lambda(\mathbf{r})$ and τ, both of which are real values. As a result, the detection probability, P_D, in (5) and the false alarm probability, P_F, in (6) are functions of $\Lambda(\mathbf{r})$ and the threshold τ.

By virtue of (5) and (6), we consider the threshold parameter τ as an independent variable and P_D and P_F as dependent variables through (5) and (6). A 3D ROC curve of (P_D,P_F,τ) extends the traditional 2D ROC curve of (P_D,P_F) by including the threshold τ to specify a third dimension in addition to the two dimensions specified by the parameters P_D and P_F used by the 2D ROC curve. In other words, the proposed 3D ROC analysis generates a 3D ROC curve plotted based on three parameters, P_D,P_F, and τ, referred to as the 3D ROC curve of (P_D,P_F,τ) from which three new 2D ROC curves can also be derived, referred to as the 2D ROC curve of (P_D,P_F), 2D ROC curve of (P_D,τ), and 2D ROC curve of (P_F,τ) where the 2D ROC curve of (P_D,P_F) turns out to be the traditional ROC curve.

Procedure for 3D ROC Curve Generation

1. Let $\mathbf{r}$ and $\delta(\mathbf{r})$ be a data sample vector and a soft decision (i.e., real-valued) classifier operating on $\mathbf{r}$, respectively.
2. Derive a normalized classifier $\delta_{\text{normalized}}(\mathbf{r})$ from $\delta(\mathbf{r})$ as
$$\delta_{\text{normalized}}(\mathbf{r}) = \frac{\delta(\mathbf{r}) - \min_{\mathbf{r}} \delta(\mathbf{r})}{\max_{\mathbf{r}} \delta(\mathbf{r}) - \min_{\mathbf{r}} \delta(\mathbf{r})}. \quad (14)$$
3. Let τ be the threshold which varies between [0,1] and a binary-valued classifier (BC), $\delta_\tau^{\text{BC}}(\mathbf{r})$, resulting from $\delta_{\text{normalized}}(\mathbf{r})$ in (13), is defined by
$$\delta_\tau^{\text{BC}}(\mathbf{r}) = \begin{cases} 1, & \text{if } \delta_{\text{normalized}}(\mathbf{r}) \geq \tau \\ 0, & \text{if } \delta_{\text{normalized}}(\mathbf{r}) < \tau \end{cases} \quad (15)$$
4. Calculate $P_D\left(\delta_\tau^{\text{BC}}(\mathbf{r})\right)$ and $P_F\left(\delta_\tau^{\text{BC}}(\mathbf{r})\right)$.
5. Generate a 3D ROC curve of (P_D, P_F, τ). (It should be noted that each threshold τ produces a point on the 3D ROC curve.)

Fig. 5 depicts a flowchart of how to generate a 3D ROC curve.

Once a 3D ROC curve is constructed, three 2D ROC curves can be obtained by projecting a 3D ROC curve on the planes of (P_D,P_F), (P_D,τ), and (P_F,τ) to yield 2D ROC curves [2D ROC curves of (P_D,P_F), 2D ROC curves of (P_D,τ), 2D ROC curves of (P_F,τ)]. Specifically, we do not have to use (5) and (6) to generate the 2D ROC curve of (P_D,P_F). By taking into account these three AUC values, we can design a new quantitative detection measure, called OD, AUC_{OD}, in a similar manner that the classification accuracy was defined in [22] by

$$-1 \leq \text{AUC}_{\text{OD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \leq 2 \quad (16)$$

where $\text{AUC}_{(F,\tau)}$ is subtracted from the sum of $\text{AUC}_{(D,\ F)}$ and $\text{AUC}_{(D,\tau)}$. This is because higher values of $\text{AUC}_{(D,\ F)}$ and $\text{AUC}_{(D,\tau)}$ indicate higher detection performance. Conversely, a lower value of $\text{AUC}_{(F,\ \tau)}$ indicates a better BKG suppression and thus a better detection performance. As a result, (16) indeed combines the values calculated by the AUC values produced by the three 2D ROC curves to produce a single quantitative value of the OD performance to evaluate each of the target detectors to be compared.

V. Utility of 3D ROC Analysis in Hyperspectral Target Detection

Despite that 3D ROC analysis has been used to evaluate target detection performance lately [8]–[22], its full utility in

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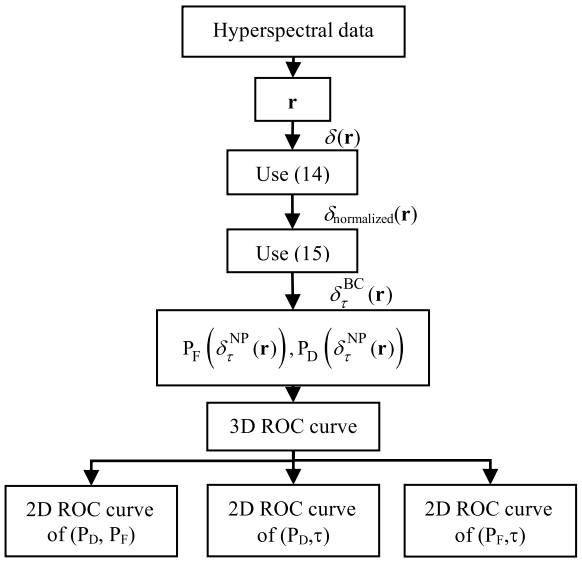


Fig. 5. Diagram of generating a 3D ROC curve.

target detection has not been explored in the past. This section discusses how a 3D ROC analysis can be further used as an evaluation tool for target detection from different aspects based on two hypotheses.

A. Effectiveness of a Detector Under Joint Hypotheses H_0 and H_1

In target detection, the 2D ROC curve of (P_D,P_F) is particularly designed to evaluate detection performance of a detector in terms of P_D versus P_F. It deals with the two hypotheses H_0 and H_1 jointly and simultaneously by using the same threshold τ, not individually and separately. So it can be only used to evaluate the effectiveness of a detector under both hypotheses. However, it often occurs that an effective detector may have a very high AUC value of a 2D ROC curve of (P_D,P_F) but may have a low P_D and a high P_F. This implies that a 2D ROC curve of (P_D,P_F) alone is not in a position to assess a detector under a single hypothesis such as the TD of a detector under H_1 or noise removal capability under H_0. The 3D ROC curve of $(P_D,P_{F,}\tau)$ provides a solution to resolve these issues by producing a 3D ROC curve which can be used to analyze all the three 2 D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) simultaneously, individually, and separately as a function of (P_D,P_F,τ).

B. Target Detectability Under a Single Hypothesis H_1

As noted in the 2D ROC curve of (P_D,P_F), the two hypotheses H_0 and H_1 actually interact with each other through the same threshold τ. Accordingly, P_D or P_F cannot be used alone to evaluate the detection performance of a detector. As will be shown in the experiments in Sections VI and VII, a detector may have a very high AUC value of (P_D,P_F), but indeed has very low AUC value of (P_D,τ). On the other hand, as also shown in the conducted experiments, it is often the case that two or more detectors generally have very close AUC values of (P_D,P_F), but their AUC values of (P_D,τ) may be quite different, a similar scenario shown in Fig. 3. In this case, judging their AUC values of (P_D,P_F) alone is difficult to conclude which detector is better than others with a negligible and tiny difference such as 0.001. The proposed 3D ROC curve allows users to produce the 2D ROC curve of (P_D,τ) independent of P_F, which can be solely used for target detection with no interfering effect from P_F.

C. Background Suppressibility Under a Single Hypothesis H_0

Hyperspectral target detection is quite different from target detection in communications/signal processing, such as radar signal processing. The crucial difference is that the former deals with target detection in images which have significant amount of BKG information, while the latter considers only signals of interest embedded in an additive noise environment governed by a random process which is generally assumed to be Gaussian. So for target detection in communications/signal processing, there is no BKG involved in the detection process. This also explains why P_{TN} (i.e., noise detection probability under H_0) in Table I has no interest at all. However, this is not true for target detection in a hyperspectral image because the image BKG has significant impact on TD, specifically anomaly detection [23]. The 2D ROC curve of (P_F,τ) derived from a 3D ROC curve of (P_D,P_F,τ) plays a key role in assessing the level of BKG being suppressed during target detection, a task that neither the 2D ROC curve of (P_D,P_F) nor the 2D ROC curve of (P_D,τ) can address.

VI. 3D ROC-Derived Quantitative Detection Measures

As discussed above, a 3D ROC curve can generate three 2D ROC curves, each of which can be used for a particular task, namely, 2D ROC curve of (P_D,P_F) used to evaluate the effectiveness of a detector, 2D ROC curve of (P_D,τ) used to dictate TD of a detector, and 2D ROC curve of (P_F,τ) used to quantify the degree of BS. However, how can we take advantage of these three 2D ROC curves to measure the detector performance becomes challenging.

A. Target Detectability

Since the AUC of a 2D ROC curve of (P_D,P_F), $\mathrm{AUC}_{(D,F)}$, and the AUC of a 2D ROC curves of (P_D,τ), $\mathrm{AUC}_{(D,\tau)}$, are indicators of how effective a detector is and how high the detection probability of a detector can achieve, respectively, the higher the values of $\mathrm{AUC}_{(D,F)}$ and $\mathrm{AUC}_{(D,\tau)}$, the better the detector. In this case, we can sum up their AUC values to define a measure, called TD of a detector as AUC_{TD}, by

$$0 \le \mathrm{AUC}_{TD} = \mathrm{AUC}_{(D,F)} + \mathrm{AUC}_{(D,\tau)} \le 2 \tag{17}$$

to quantitatively calculate the effectiveness and TD of a detector.

B. Background Suppressibility

On the other hand, since the AUC of a 2D ROC curve of (P_F,τ), $\mathrm{AUC}_{(F,\tau)}$, uses P_F as an indicator to describe the BS, the smaller the value of $\mathrm{AUC}_{(F,\tau)}$, the better the detector.

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So, factoring $\mathrm{AUC}_{(\mathrm{F},\tau)}$ into a detector yields a measure, called BS of a detector, $\mathrm{AUC}_{\mathrm{BS}}$, defined by

$$-1 \le \mathrm{AUC}_{\mathrm{BS}} = \mathrm{AUC}_{(\mathrm{D,F})} - \mathrm{AUC}_{(\mathrm{F},\tau)} \le 1. \tag{18}$$

C. Joint Target Detectability With Background Suppressibility

In the definitions of TD and BS, $\mathrm{AUC}_{(\mathrm{D,F})}$ is used to measure the effectiveness of a detector when P_{D} and P_{F} use the same threshold τ for their joint performance without accounting in TD and BS separately. To factor $\mathrm{AUC}_{(\mathrm{F},\tau)}$ into $\mathrm{AUC}_{(\mathrm{D},\tau)}$, we need to define a measure to evaluate JTDBS of a detector. In this case, there are three different ways to jointly consider TD and BS.

1) TD in BKG: The first JTDBS measure is to consider TD and BS working jointly with different values of the threshold τ as an independent parameter. Since P_{F} is caused by the probability of misdetecting noise as a signal, this error probability should be subtracted from P_{D}. To take care of this effect, a measure, factoring P_{F} into P_{D}, to be called TD-BS, can be defined as $\mathrm{AUC}_{\mathrm{TD-BS}}$ by

$$-1 \le \mathrm{AUC}_{\mathrm{TD\text{-}BS}} = \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)} \le 1. \tag{19}$$

2) Overall Detection Probability: Another important JTDBS measure is derived from a classification point of view. It is called ODP defined by

$$0 \le \mathrm{AUC}_{\mathrm{ODP}} = \mathrm{AUC}_{(\mathrm{D},\tau)} + (1 - \mathrm{AUC}_{(\mathrm{F},\tau)}) \le 2 \tag{20}$$

where $\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $1 - \mathrm{AUC}_{(\mathrm{F},\tau)}$ can be reinterpreted as the accuracy of TD and accuracy of BKG detectability, respectively. As a result, $\mathrm{AUC}_{\mathrm{ODP}}$ turns out to be the same measure in correspondence to the commonly used in classification, called overall accuracy (OA) in classification with the sample ratio factored into OA to account for the weight assigned to each class. So (20) is indeed equivalent to (19) and both are identical with a difference by the constant 1.

3) Medical Diagnosis Probability: Interestingly, (20) can be further reinterpreted from a medical diagnosis point of view and defined as MDP by

$$\begin{aligned} 0 \le \mathrm{AUC}_{\mathrm{MDP}} &= \mathrm{AUC}_{(\mathrm{D},\tau)} + \left(1 - \mathrm{AUC}_{(\mathrm{F},\tau)}\right) \\ &= \mathrm{AUC}_{\mathrm{SS}} + \mathrm{AUC}_{\mathrm{SF}} \le 2 \end{aligned} \tag{21}$$

where $\mathrm{AUC}_{\mathrm{SS}}$=$\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $\mathrm{AUC}_{\mathrm{SF}} = \mathrm{AUC}_{\mathrm{TN}} = 1 - \mathrm{AUC}_{(\mathrm{F},\tau)}$ are known as sensibility (SS) or TP and specificity (SF) or true negative (TN). So MDP is identical to ODP.

D. Signal-to-Noise Probability Ratio

The last but most effective detection measure comes from a similar idea that is widely used in communications/signal processing, signal-to-noise ratio (SNR). It is called signal-to-noise probability ratio (SNPR) defined by

$$0 \le \mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(\mathrm{D},\tau)}}{\mathrm{AUC}_{(\mathrm{F},\tau)}} \tag{22}$$

where the false alarm probability P_{F} can be assumed to be caused by noise.

Finally, $\mathrm{AUC}_{\mathrm{OD}}$ in (16) is particularly designed to combine all the three measures, $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, and $\mathrm{AUC}_{(\mathrm{F},\tau)}$, together into one measure by subtracting $\mathrm{AUC}_{(\mathrm{F},\tau)}$ from $\mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{(\mathrm{D},\tau)}$ to yield an overall indicator for a detector.

To the author's best knowledge, the six measures, $\mathrm{AUC}_{\mathrm{TD}}$ in (17), $\mathrm{AUC}_{\mathrm{BS}}$ in (18), $\mathrm{AUC}_{\mathrm{TD-BS}}$ in (19), $\mathrm{AUC}_{\mathrm{SNPR}}$ in (20), $\mathrm{AUC}_{\mathrm{ODP}}$ in (21), and $\mathrm{AUC}_{\mathrm{MDP}}$ in (22), along with $\mathrm{AUC}_{\mathrm{OD}}$ in (16) are the first ever proposed in hyperspectral target detection or anomaly detection as indicators to measure detection performance of a detector from all-around aspects.

E. 2D/3D ROC Analysis Software

To facilitate the use of 2D/3D ROC analysis, MATLAB-based software codes developed for calculating AUC and plotting ROC are made available at the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, https://wiki.umbc.edu/display/rssipl/10.+Download where the functions cal_AUC.m and cal_pdpf.m are used to calculate AUC and normalized AUC based on (7), (9), and (13) and the function plot_3DROC.m is used to plot 3D ROC and 2D ROC curves.

As final comments, it should also be noted that the experiments to be conducted in Sections VII and VIII were specifically designed to demonstrate how the issues described above can be addressed.

1) First, theoretical derivations are developed to correctly generate 2D ROC curve of $(P_{\mathrm{D}},P_{\mathrm{F}})$ with an algorithm provided at the end of Section III.
2) Various types of new quantitative detection measures are designed and developed. For example, $\mathrm{AUC}_{(\mathrm{D,F})}$ and $\mathrm{AUC}_{\mathrm{TD}}$ in (17) to evaluate effectiveness in detection performance of a detector, $\mathrm{AUC}_{\mathrm{BS}}$ in (18) to assess BS of a detector, $\mathrm{AUC}_{\mathrm{TD-BS}}$ in (19) and $\mathrm{AUC}_{\mathrm{ODP}}$ in (20) to analyze detection of targets embedded in BKG, and $\mathrm{AUC}_{\mathrm{SNPR}}$ in (22) to measure the ratio of P_{D} to P_{F}, SNPR.
3) Most importantly, using a 3D ROC curve can avoid the issues arising in generating 2D ROC curve of $(P_{\mathrm{D}},P_{\mathrm{F}})$ discussed in Section III. This is because a 3D ROC curve is generated by three parameters, $P_{\mathrm{D}},P_{\mathrm{F}}$, and τ and the 2D ROC curve of $(P_{\mathrm{D}},P_{\mathrm{F}})$ can be obtained directly by simply projecting the 3D ROC curve on the plane of $(P_{\mathrm{D}},P_{\mathrm{F}})$ with no need of using (11) and (12). Nevertheless, a normalization is still also needed.
4) To illustrate the utility of 3D ROC analysis along with the newly designed quantitative detection measures defined above, target detection is performed on two types of targets, prior known targets as prior target detection in Section VII and anomalous targets as anomaly detection in Section VIII.

VII. Experiments for Prior Target Detection

To demonstrate the utility of 3D ROC analysis for target detection, this section considers prior target detection where the targets of interest were provided as known targets by prior knowledge and the target detector to be studied is the CEM developed in [24]–[39] along with its two recent variants, H-CEM [30] and E-CEM [31] with their parameter settings specified in Table II.

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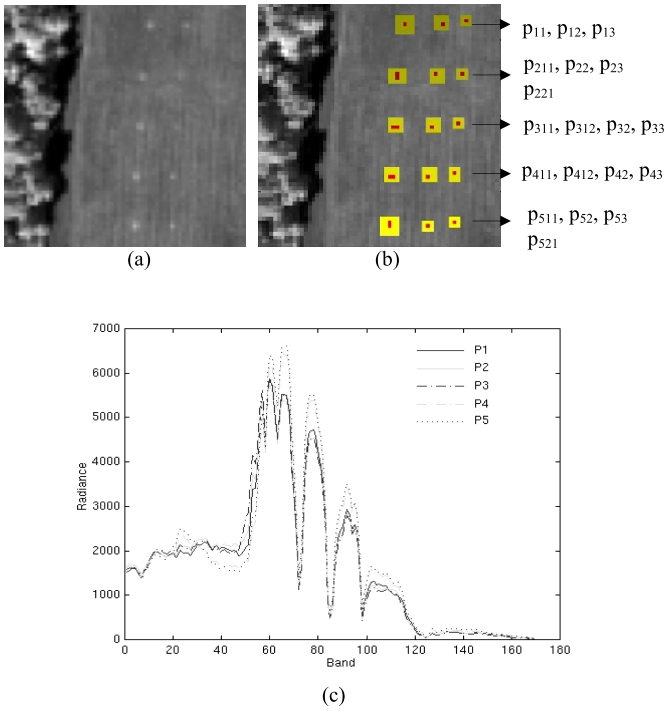


Fig. 6. (a) HYDICE panel scene which contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels. (c) Spectra of $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$.

TABLE II
PARAMETER SPECIFICATIONS USED BY H-CEM AND E-CEM

detectors	parameters used by detectors
H-CEM	$\alpha = 200$
E-CEM	number of used scanning windows = $n_w = 4$ windows size = [10, 42, 95, 169] $n_{\text{CEM}} = 6$

The image scene to be used for experiments shown in Fig. 6 was acquired by the airborne HYDICE sensor in August 1995 from a flight altitude of 10 000 ft. This scene has been studied extensively by many reports such as [19], [23], [29], and [32].

There are 15 square panels with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively, shown in Fig. 6(a). Due to the approximately 1.56-m ground sampling distance, each of the panels in the first column except the first row contains two panel pixels highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 6(b) by ground truth. All the 11 remaining panels in Fig. 6(b) contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Therefore, there are a total of 19 panel pixels. Fig. 6(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. Fig. 6(c) plots the five panel spectral signatures, $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$ obtained from Fig. 6(b), where the ith panel signature, denoted by $\mathbf{p}_i$, was generated by averaging the red panel center pixels in row i. These five signatures were then used as the desired prior target signatures for CEM.

Fig. 7(a)–(c) shows the detection maps of the panels in rows 1–5 produced by CEM, H-CEM, and E-CEM, respectively. By visual inspection of Fig. 7, E-CEM and C-CEM performed very comparably and clearly better in detecting all 19 R panel pixels in Fig. 7(b) and (c) than CEM in Fig. 7(a) where H-CEM and E-CEM detected all the three invisible subpanel pixels, p_{13}, p_{22}, and p_{33} in the third column in Fig. 7(b) and (c) but CEM could barely detect them in Fig. 7(a).

However, the qualitative inspection of Fig. 7 does not quantify how much closeness is observed between H-ECM and E-CEM and how much improvement over CEM. The developed 2D/3D ROC analysis provides an effective means to evaluate their detection performances quantitatively. In doing so, we need to produce discrete 2D ROC curves of $(P_{\text{D}}, P_{\text{F}})$ of the detection maps in Fig. 7 where $\delta^{\text{NP}(1)}_{\text{normalized}}(\mathbf{r})$ specified by (7) and $\delta^{\text{NP}(0)}_{\text{normalized}}(\mathbf{r})$ specified by (9) are used to threshold real-valued of detection maps to their discrete values.

A. Discrete ROC Curves Generated by $\delta^{NP(1)}_{normalized}(\mathbf{r})$ in (7)

As an illustrative example, Fig. 8(a) shows a common scenario that occurs in most reported research works using (7) to threshold a detector to generate a 2D ROC curve with the original A_z shaded by gray specified by $a_{\tau=1} > 0$ and $a_{\tau=0} = 1$, and $b_{\tau=1} = 0$ and $b_{\tau=0} = 1$. In this case, the discrete-valued 2D ROC curve is generated by only a finite number of data samples if data sample values are used as thresholds or a finite number of endpoints of step-size-partitioned intervals used as thresholds. When the AUC is calculated, it integrates the 2D ROC curve with P_{F} from 0 to 1, that is, the sum of the gray area A_z and the area $(1 - a_{\tau=1})$ underneath the gray-shaded A_z. Theoretically speaking, this is incorrect because there are no values of P_{F} corresponding to the values of P_{D} lower than $a_{\tau=1}$ which is already obtained by $P_{\text{F}} = 0$. So the correct AUC should be the one normalizing A_z by $(1 - a_{\tau=1})$ as shown in Fig. 8(b) by $A_z^{\text{normalized}} = A_z/(1 - a_{\tau=1})$ via (13).

So using $\delta^{\text{NP}(1)}_{\text{normalized}}(\mathbf{r})$ in (7) to threshold the detection maps in Fig. 7 produced Figs. 9(a)–(c) and 10(a)–(c) which showed their discrete 2D ROC curves of $(P_{\text{D}}, P_{\text{F}})$ and their corresponding normalized counterparts using the uniform step size of $\Delta = 0.01$ and data sample values, respectively, where the curves are zoomed-in in very small ranges to see their differences. As we can see that the generated 2D ROC curves in Figs. 9 and 10 are similar to those in Fig. 8 with $a_{\tau=1} > 0$ and A_z normalized by $(1 - a_{\tau=1})$. Figs. 11 and 12 also show the 3D ROC curves along with their corresponding three 2D ROC curves using (7) to threshold the detection maps in Fig. 7.

B. Discrete ROC Curves Generated by $\delta^{NP(0)}_{normalized}(\mathbf{r})$ in (9)

In analogy with using $\delta^{\text{NP}(1)}_{\text{normalized}}(\mathbf{r})$ in (7) to produce Fig. 8, we also used $\delta^{\text{NP}(0)}_{\text{normalized}}(\mathbf{r})$ in (9) to threshold a detector to generate a discrete 2D ROC curve with the original A_z shaded by gray specified by $a_{\tau=1} = 0$ and $a_{\tau=0} = 1$ and $b_{\tau=1} = 0$ and $b_{\tau=0} < 1$ as shown in Fig. 13(a). So the correct AUC should be the one normalizing A_z by $(1 - a_{\tau=1})$, as shown in Fig. 13(b) by $A_z^{\text{normalized}} = \text{A}_z/b_{\tau=1}$ via (13).

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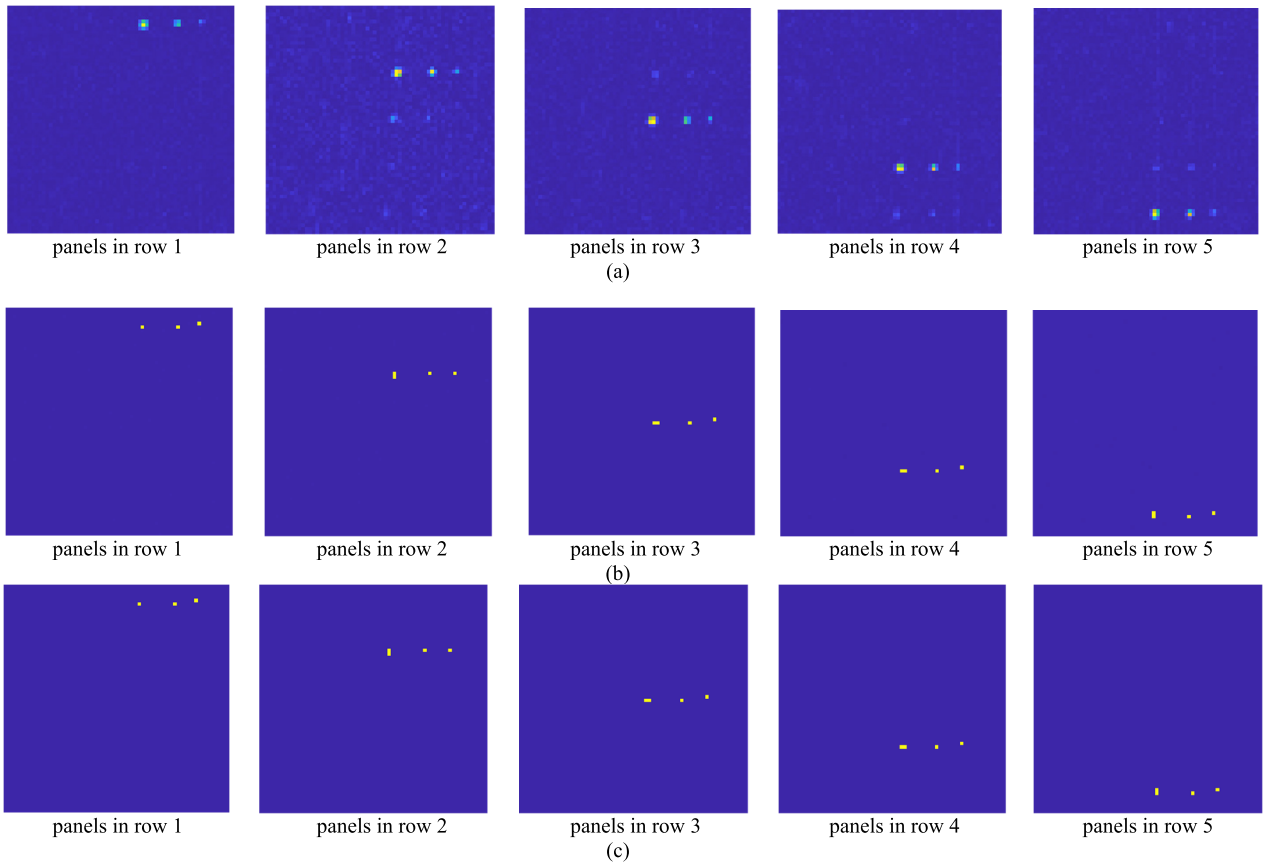


Fig. 7. Detection maps of panels in five rows in Fig. 6 by CEM, H-CEM, and E-CEM. (a) Detection maps of CEM. (b) Detection maps of H-CEM. (c) Detection maps of E-CEM.

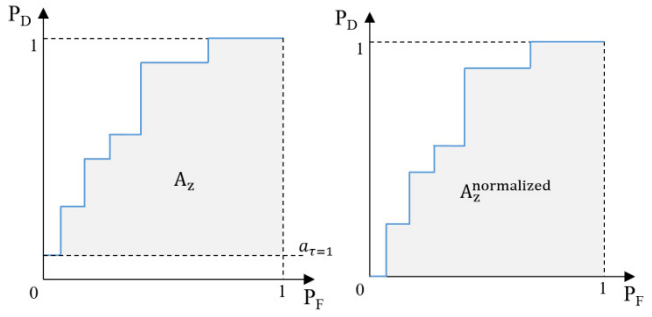


Fig. 8. Original discrete ROC curve and its normalized ROC curve using (7).

Figs. 14(a)–(c) and 15(a)–(c) show the discrete 2D ROC curves of (P_D, P_F) using data sample values with normalization, respectively, where (9) was used to threshold the detection maps in Fig. 7.

As shown, the discrete 2D ROC curves in Figs. 14 and 15 are similar to those in Fig. 13 with $b_{\tau=0} < 1$ and A_z normalized by $b_{\tau=0}$. Figs. 16 and 17 also shows the 3D ROC curves along with their corresponding three 2D ROC curves using (9) to threshold the detection maps in Fig. 7.

To further conduct a quantitative analysis on the 3D ROC curves Tables III and IV also tabulate the AUC values of $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, $\text{AUC}_{(F,\tau)}$, and AUC_{TD} in (17), AUC_{BS} in (18), AUC_{TD-BS} in (19), AUC_{ODP} in (20), AUC_{SNPR} in (22), and AUC_{OD} in (16), all of which were calculated by (7) from Figs. 9 to 12 using step-size and data sample values as threshold values where the best cases among the three detectors are boldfaced. Also, except the case of detecting panel pixels in row 1 with $a_{\tau=1} = 0.3333$, all other cases had $a_{\tau=1} = 0.25000$ and $a_{\tau=0} = 1$, and $b_{\tau=1} = 0$ and $b_{\tau=0} = 1$. Apparently, on the whole, H-CEM and E-CEM performed significantly better than CEM. As noted in Fig. 7 and the AUC values of (P_D, P_F), H-CEM and E-CEM performed exactly the same with $\text{AUC}_{(D,F)} = 1$ and CEM has its AUC values of (P_D, P_F) around 0.999 from 0.999. So we only evaluate these three target detectors solely based on the AUC values of (P_D, P_F), and we may easily conclude that these three detectors performed nearly the same within a tolerance range of 10^{-3}. However, the quantification results in Tables III and IV revealed that this was not true if the other six detection measures, $\text{AUC}_{(D,\tau)}$ $\text{AUC}_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TD-BS}, AUC_{OD}, AUC_{ODP}, and AUC_{SNPR}, were taken into account for further evaluation. As a matter of fact, E-CEM indeed performed slightly better than H-CEM almost in all the six detection measures regardless of whether A_z was normalized, particularly, for the cases using data sample values as thresholds in Table IV.

Similarly, the same conclusion can be drawn for 3D ROC curves where the AUC values were calculated in Tables V and VI (9) using step-size and data sample values

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TABLE III

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR FIVE ROWS OF DETECTION RESULTS USING (7) AND UNIFORM STEP SIZE, $\Delta = 0.01$

	AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
$\mathbf{p_1}$	A_z	CEM	0.3333/1	0/1	0.999267	0.590000	0.009009	1.589267	0.990258	0.580991	1.580991	1.580258	65.49006
		H-CEM			**1**	**0.996667**	**0.005000**	**1.996667**	**0.995000**	**0.994667**	**1.994667**	**1.991667**	**199.3334**
		E-CEM			**1**	**0.996667**	**0.005000**	**1.996667**	**0.995000**	**0.994667**	**1.994667**	**1.991667**	**199.3334**
	$A_z^{normalized}$	CEM			0.998901	0.385000	0.009009	1.383901	0.375991	0.375991	1.375991	1.374891	42.73504
		H-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
		E-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
$\mathbf{p_2}$	A_z	CEM	0.2500/1	0/1	0.999817	0.821250	0.017224	1.821067	0.982593	0.804026	1.804026	1.803843	47.68056
		H-CEM			**1**	**0.996250**	**0.00500**	**1.996250**	**0.995000**	**0.995750**	**1.995750**	**1.991250**	**199.2500**
		E-CEM			**1**	**0.996250**	**0.00500**	**1.996250**	**0.995000**	**0.995750**	**1.995750**	**1.991250**	**199.2500**
	$A_z^{normalized}$	CEM			0.999756	0.761667	0.017224	1.761423	0.982532	0.744447	1.744447	1.744198	44.22126
		H-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
		E-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
$\mathbf{p_3}$	A_z	CEM	0.2500/1	0/1	0.999695	0.728750	0.011219	1.728445	0.988476	0.717531	1.717531	1.717225	64.95676
		H-CEM			**1**	**0.996250**	**0.005000**	**1.996250**	**0.995000**	**0.991250**	**1.991250**	**1.991250**	**199.2500**
		E-CEM			**1**	**0.996250**	**0.005000**	**1.996250**	**0.995000**	**0.991250**	**1.991250**	**1.991250**	**199.2500**
	$A_z^{normalized}$	CEM			0.999593	0.638333	0.011219	1.637926	0.988374	0.627114	1.627114	1.626707	56.89749
		H-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
		E-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
$\mathbf{p_4}$	A_z	CEM	0.2500/1	0/1	0.999817	0.761250	0.010887	1.761067	0.988930	0.750363	1.750363	1.750180	69.92284
		H-CEM			**1**	**0.996250**	**0.005000**	**1.996250**	**0.995000**	**0.991250**	**1.991250**	**1.991250**	**199.2500**
		E-CEM			**1**	**0.996250**	**0.005000**	**1.996250**	**0.995000**	**0.991250**	**1.991250**	**1.991250**	**199.2500**
	$A_z^{normalized}$	CEM			0.999756	0.681667	0.010887	1.681423	0.988869	0.670780	1.670780	1.670535	62.61293
		H-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
		E-CEM			1	0.995000	0.005000	1.995000	0.995000	0.990000	1.990000	1.990000	199.0000
$\mathbf{p_5}$	A_z	CEM	0.2500/1	0/1	0.999756	0.686250	0.010887	1.686006	0.988869	0.675363	1.675363	1.675119	63.03389
		H-CEM			**1**	**0.996250**	0.005007	**1.996250**	0.994993	0.991243	1.991243	1.991243	198.9714
		E-CEM			**1**	**0.996250**	**0.005000**	**1.996250**	**0.995000**	**0.991250**	**1.991250**	**1.991250**	**199.2500**
	$A_z^{normalized}$	CEM			0.999674	0.581667	0.010887	1.581341	0.988787	0.570780	1.570780	1.570454	53.42766
		H-CEM			1	0.995000	0.005007	1.995000	0.994993	0.989993	1.989993	1.989993	198.7217
		E-CEM			1	0.995000	**0.005000**	1.995000	**0.995000**	0.990000	1.990000	1.990000	199.0000

TABLE IV

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR FIVE ROWS OF DETECTION RESULTS USING (7) AND DATA SAMPLE VALUES

	AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
$\mathbf{p_1}$	A_z	CEM	0.3333/1	0/1	0.999267	0.595313	0.008401	1.589458	0.990866	0.586912	1.586912	1.586179	70.86216
		H-CEM			**1**	0.999839	0.003753	1.999839	0.996247	0.996086	1.996086	1.996087	266.4106
		E-CEM			**1**	**0.999990**	**0.001122**	**1.999990**	**0.998878**	**0.998868**	**1.998868**	**1.998869**	**891.2567**
	$A_z^{normalized}$	CEM			0.998901	0.392970	0.008401	1.391871	0.990500	0.384569	1.384569	1.383469	46.77657
		H-CEM			**1**	0.999759	0.003753	1.99759	0.995000	0.996006	1.996006	1.996007	266.3893
		E-CEM			**1**	0.999985	**0.001122**	1.999985	0.995000	0.998863	1.998863	1.998864	891.2522
$\mathbf{p_2}$	A_z	CEM	0.2500/1	0/1	0.999817	0.832485	0.016907	1.832302	0.982910	0.815578	1.815578	1.815395	49.23907
		H-CEM			**1**	0.999717	0.003599	1.999717	0.996401	0.996118	1.996118	1.996118	277.7763
		E-CEM			**1**	**0.999987**	**0.001153**	**1.999987**	0.995000	**0.998834**	**1.998834**	**1.998835**	**867.2914**
	$A_z^{normalized}$	CEM			0.999756	0.776647	0.016907	1.776493	0.982849	0.759740	1.759740	1.759496	45.93642
		H-CEM			**1**	0.999622	0.003599	1.999622	0.996401	0.996023	1.996023	1.996023	277.7499
		E-CEM			**1**	0.999983	**0.001153**	1.999983	**0.998847**	0.998830	1.998830	1.998831	867.2879
$\mathbf{p_3}$	A_z	CEM	0.2500/1	0/1	0.999695	0.765813	0.010802	1.765500	0.988893	0.755011	1.755011	1.754706	70.89548
		H-CEM			**1**	0.999656	0.005620	1.999656	0.994380	0.994036	1.994036	1.994035	177.8747
		E-CEM			**1**	**0.999988**	**0.001576**	**1.999988**	**0.998424**	**0.998412**	**1.998412**	**1.998413**	**634.5102**
	$A_z^{normalized}$	CEM			0.999593	0.687751	0.010802	1.687344	0.988791	0.676949	1.676949	1.676542	63.66886
		H-CEM			**1**	0.999541	0.005620	1.999541	0.994380	0.993921	1.993921	1.993920	177.8543
		E-CEM			**1**	**0.999984**	**0.001576**	1.999984	**0.998424**	0.998408	1.998408	1.998409	634.5076
$\mathbf{p_4}$	A_z	CEM	0.2500/1	0/1	0.999817	0.814910	0.010377	1.814727	0.989440	0.804533	1.804533	1.804350	78.5304
		H-CEM			**1**	0.999681	0.005310	1.999681	0.994690	0.994371	1.994371	1.994371	188.2638
		E-CEM			**1**	**0.999988**	**0.002126**	**1.999988**	**0.997874**	**0.997862**	**1.997862**	**1.997862**	**470.3612**
	$A_z^{normalized}$	CEM			0.999756	0.753214	0.010377	1.752971	0.989397	0.742837	1.742837	1.742593	72.58495
		H-CEM			**1**	0.999574	0.005310	1.999574	0.994690	0.994264	1.994264	1.994264	188.2437
		E-CEM			**1**	0.999984	**0.002126**	1.999984	**0.997874**	0.997858	1.997858	1.997858	470.3594
$\mathbf{p_5}$	A_z	CEM	0.2500/1	0/1	0.999756	0.737585	0.010322	1.737341	0.989493	0.727263	1.727263	1.727019	71.45757
		H-CEM			**1**	0.999357	0.008521	1.999357	0.991479	0.990836	1.990836	1.990836	117.2817
		E-CEM			**1**	**0.999956**	**0.001163**	**1.999956**	**0.998837**	**0.998793**	**1.998793**	**1.998793**	**859.8074**
	$A_z^{normalized}$	CEM			0.999674	0.650114	0.010322	1.649788	0.989352	0.639792	1.639792	1.639466	62.98334
		H-CEM			**1**	0.999143	0.008521	1.999143	0.991479	0.990622	1.990622	1.990622	117.2565
		E-CEM			**1**	0.999941	**0.001163**	1.999941	**0.998837**	0.998778	1.998778	1.998778	859.7945

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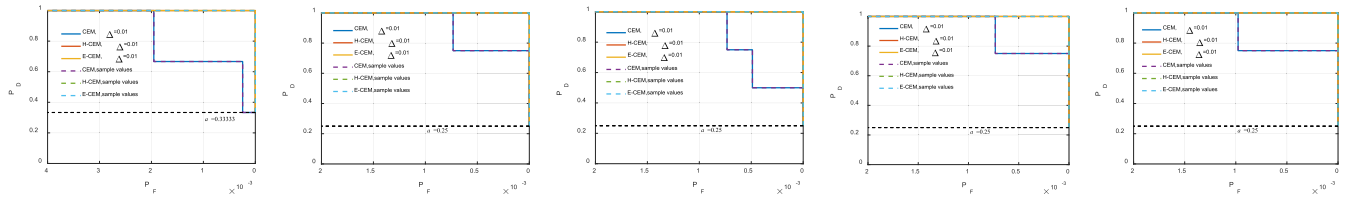


Fig. 9. 2D ROC curves of (P_D, P_F) produced in Fig. 8 by CEM, H-CEM, and E-CEM using (7) by data sample values and uniform step size, $\Delta = 0.01$.

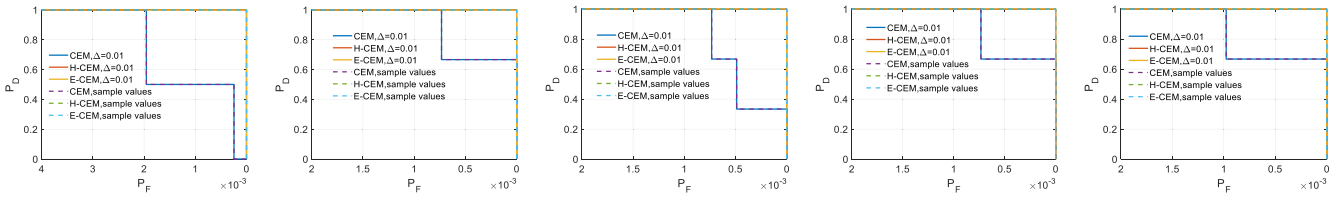


Fig. 10. Normalized 2D ROC curves of (P_D, P_F) produced by CEM, H-CEM, and E-CEM using (7) by data sample values and uniform step size, $\Delta = 0.01$.

TABLE V

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR FIVE ROWS OF DETECTION RESULTS USING (9) AND UNIFORM STEP SIZE, $\Delta = 0.01$

	AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
$\mathbf{p_1}$	A_z	CEM	0/1	0/0.9998	0.999023	0.588333	0.009008	1.587356	0.990015	0.579325	1.579325	1.578348	65.31228
		H-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
		E-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
	$A_z^{\text{normalized}}$	CEM			0.999267	0.588333	0.009010	1.587600	0.990257	0.579323	1.579323	1.578592	65.29778
		H-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
		E-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
$\mathbf{p_2}$	A_z	CEM	0/1	0/0.9998	0.999572	0.820000	0.017223	1.819572	0.982349	0.802777	1.802777	1.802349	47.61075
		H-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
		E-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
	$A_z^{\text{normalized}}$	CEM			0.999817	0.820000	0.017227	1.819817	0.982590	0.802773	1.802773	1.802594	47.59970
		H-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
		E-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
$\mathbf{p_3}$	A_z	CEM	0/1	0/0.9998	0.999450	0.727500	0.011218	1.726950	0.988232	0.716282	1.716282	1.715732	64.85113
		H-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
		E-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
	$A_z^{\text{normalized}}$	CEM			0.999694	0.727500	0.011221	1.727194	0.988473	0.716279	1.716279	1.715976	64.83379
		H-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
		E-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
$\mathbf{p_4}$	A_z	CEM	0/1	0/0.9998	0.999572	0.760000	0.010886	1.759572	0.988686	0.749114	1.749114	1.748686	69.81444
		H-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	1.990001	1.989757	199.0398
		E-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
	$A_z^{\text{normalized}}$	CEM			0.999817	0.760000	0.010889	1.759817	0.988928	0.749111	1.749111	1.748931	69.79521
		H-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
		E-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000
$\mathbf{p_5}$	A_z	CEM	0/1	0/0.9998	0.999511	0.685000	0.010886	1.684511	0.988625	0.674114	1.674114	1.673625	62.92486
		H-CEM			0.999756	**0.995000**	0.005006	1.994756	0.992554	0.989994	1.989994	1.989750	198.7615
		E-CEM			0.999756	**0.995000**	**0.004999**	1.994756	0.994757	**0.990001**	**1.990001**	1.989757	**199.0398**
	$A_z^{\text{normalized}}$	CEM			0.999756	0.685000	0.010889	1.684756	0.988857	0.674111	1.674111	1.673870	62.90752
		H-CEM			**1**	**0.995000**	0.005007	**1.995000**	0.994993	0.989993	1.989993	1.989994	198.7218
		E-CEM			**1**	**0.995000**	0.005000	**1.995000**	**0.995000**	0.990000	1.990000	**1.990001**	1990000

as threshold values where the best cases among the three detectors are boldfaced. Interestingly, unlike cases in Tables III and IV, using (9) produced different values of $a_{\tau=1}$ and $a_{\tau=0}$, and $b_{\tau=1}$ and $b_{\tau=0}$. When using step size as thresholding, all the cases had $a_{\tau=1} = 0$ and $a_{\tau=0} = 1$, and $b_{\tau=1} = 0$, but $b_{\tau=0} = 0.9998$ not equal to 1. However, if data sample values were used as threshold values, $a_{\tau=1} = 0$ and $a_{\tau=0} = 1$, and $b_{\tau=1} = 0$ and $b_{\tau=0} = 0.9998$ for all the cases of detecting panel pixels in rows 1–4. But for the case of detecting panel pixels in row 5, the result was reversed, that is, $a_{\tau=1} = 0.25000$, $a_{\tau=0} = 1$, and $b_{\tau=1} = 0$, but $b_{\tau=0} = 1$. This is quite different from Tables III and IV. This also indicated that a detector using $\delta^{\text{NP}(1)}_{\text{normalized}}(\mathbf{r})$ and $\delta^{\text{NP}(0)}_{\text{normalized}}(\mathbf{r})$ specified by (7) and (9), respectively, indeed, made a difference.

Comparing the results in Tables III–V, it is very clear that using data sample values as threshold values can assess

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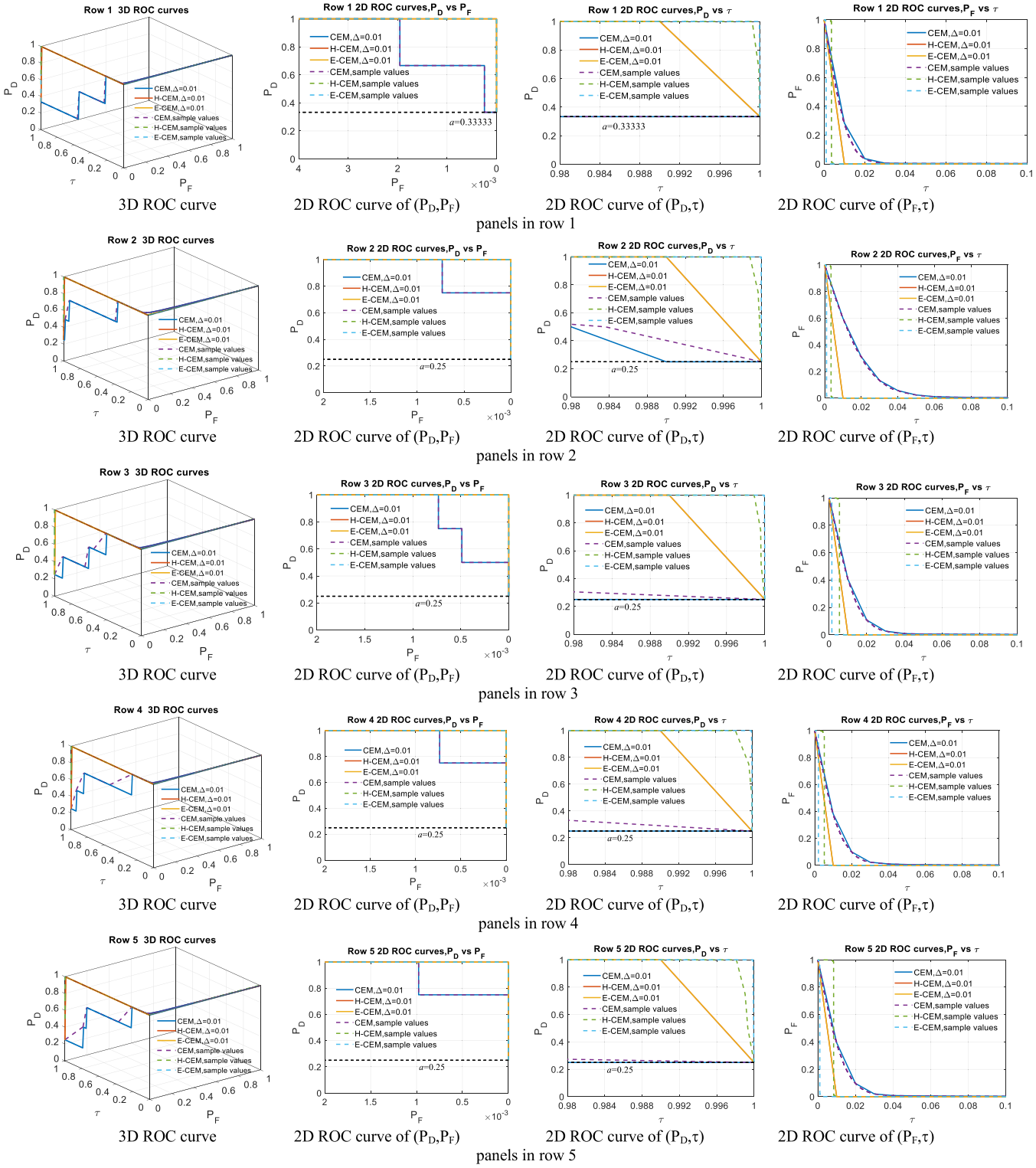


Fig. 11. 3D ROC curves along with their generated three 2D ROC curves using (7) by data sample values and uniform step size, $\Delta = 0.01$.

detection performance more accurately and reveal subtle difference between H-CEM and E-CEM where their performances were so close that tiny differences can be uncovered using step size as thresholding. For example, for detection of panel pixels in rows from 1 to 4, H-CEM and E-CEM produced identical results in Table III using step size as thresholding. However, in fact, they performed slightly differently if data sample values were used as threshold values in Table IV. Similarly, it was also true for Tables V and VI.

An interesting observation is worth noting. In evaluation of detection performance, if we only paid attention to $\mathrm{AUC}_{(\mathrm{D,F})}$, their values produced by CEM, H-CEM, and

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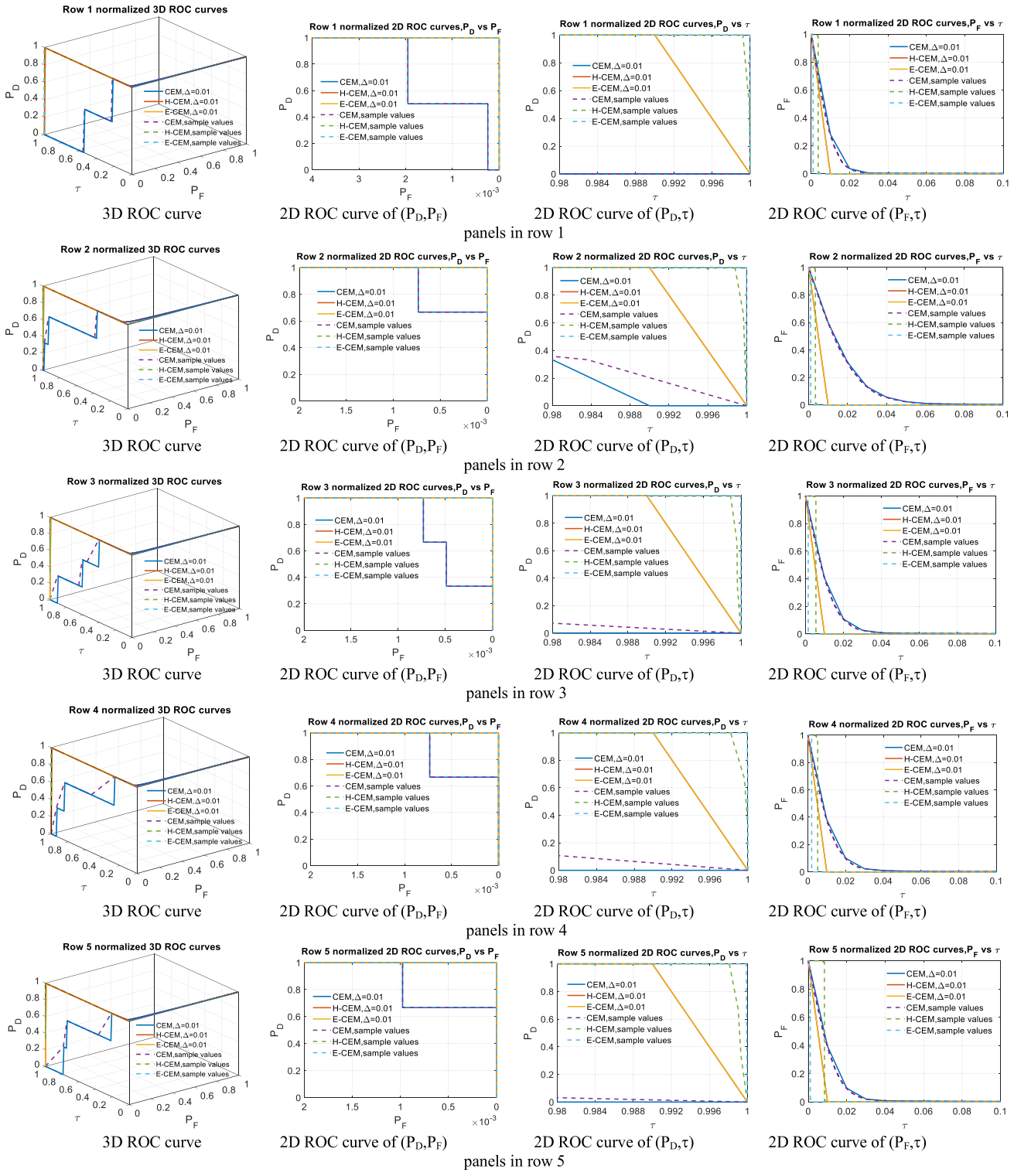


Fig. 12. Normalized 3D ROC curves along with their generated three normalized 2D ROC curves using (7) by data sample values and uniform step size, $\Delta = 0.01$.

E-CEM were very close within the range of 0.001. This may lead to a conclusion that these three CEM-based detectors performed nearly the same. As a matter of fact, if we examine the values of their $\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $\mathrm{AUC}_{(\mathrm{F},\tau)}$, their differences were actually appreciable. Specifically, $\mathrm{AUC}_{\mathrm{SNPR}}$ provided significant and valuable information where E-CEM was many times higher than CEM and 3–4 times higher than H-CEM from Table VI. Accordingly, it can be easily concluded that E-CEM outperformed H-CEM in all aspects. This indicates that when we come to assess performance of a detector, $\mathrm{AUC}_{\mathrm{SNPR}}$ alone may provide valuable information in terms of TD and BS, respectively. Specifically, as shown in

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TABLE VI

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR FIVE ROWS OF DETECTION RESULTS USING (9) AND DATA SAMPLE VALUES

	AUCs	Detector	$a_{\tau=1}/a_{\tau=0}$	$b_{\tau=0}/b_{\tau=1}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
$\mathbf{p_1}$	A_z	CEM	0/1	0/0.9998	0.999023	0.502596	0.008225	1.501619	0.990798	0.494371	1.494371	1.493393	61.10590
		H-CEM			0.999756	**0.834097**	0.003623	1.833853	0.996133	0.830474	1.830474	1.830230	230.2227
		E-CEM			0.999756	0.833763	**0.000999**	1.833519	0.998757	**0.832764**	**1.832764**	1.832519	**834.5976**
	$A_z^{\text{normalized}}$	CEM			0.999267	0.502596	0.008227	1.501863	0.991040	0.494369	1.494369	1.493637	61.09104
		H-CEM			**1**	**0.834097**	0.003624	**1.834097**	0.996376	0.830473	1.830473	1.830475	230.1592
		E-CEM			**1**	0.833763	**0.000999**	1.833763	**0.999001**	**0.832764**	**1.832764**	**1.832763**	**834.5976**
$\mathbf{p_2}$	A_z	CEM	0/1	0/0.9998	0.999572	0.796535	0.016697	1.796107	0.982875	0.779838	1.779838	1.779410	47.70528
		H-CEM			0.999756	**0.875586**	0.003466	1.875342	0.996290	0.872120	1.872120	1.871876	252.6215
		E-CEM			0.999756	0.875175	**0.001030**	1.874931	**0.998726**	**0.874145**	**1.874145**	1.873900	**849.6845**
	$A_z^{\text{normalized}}$	CEM			0.999817	0.796535	0.016702	1.796352	0.983115	0.779833	1.779833	1.779655	47.69100
		H-CEM			**1**	**0.875586**	0.003467	**1.875586**	0.996533	0.872119	1.872119	1.872120	252.5486
		E-CEM			**1**	0.875175	**0.001030**	**1.875586**	0.998970	**0.874145**	**1.874145**	**1.874144**	**849.6845**
$\mathbf{p_3}$	A_z	CEM	0/1	0/0.9998	0.999450	0.679804	0.010641	1.679254	0.988809	0.669163	1.669163	1.668613	63.88535
		H-CEM			0.999756	**0.875763**	0.005497	1.875519	0.992063	0.870266	1.870266	1.870021	159.3165
		E-CEM			0.999756	0.875287	**0.001453**	1.874043	0.998303	**0.873834**	**1.873834**	**1.873590**	**602.3999**
	$A_z^{\text{normalized}}$	CEM			0.999694	0.679804	0.010644	1.679498	0.989050	0.669160	1.669160	1.668857	63.86734
		H-CEM			**1**	**0.875763**	0.005498	**1.875763**	0.994502	0.870265	1.870265	1.870265	159.2876
		E-CEM			**1**	0.875287	0.001454	1.875287	**0.998546**	0.873833	1.873833	1.873834	601.9856
$\mathbf{p_4}$	A_z	CEM	0/1	0/0.9998	0.999572	0.718000	0.010227	1.717572	0.989345	0.707773	1.707773	1.707345	70.20632
		H-CEM			0.999756	0.875584	0.005187	1.875341	0.994569	0.870397	1.870397	1.870153	168.8035
		E-CEM			0.999756	**0.875425**	**0.002003**	**1.895181**	0.997752	**0.873422**	**1.873422**	1.873177	**437.0569**
	$A_z^{\text{normalized}}$	CEM			0.999817	0.718000	0.01023	1.717817	0.989587	0.707770	1.707770	1.707590	70.18573
		H-CEM			**1**	0.875584	0.005188	1.875584	0.994812	0.870396	1.870396	1.870397	168.7710
		E-CEM			**1**	**0.875425**	0.002004	1.875425	**0.997996**	0.873421	1.873421	**1.873422**	436.8388
$\mathbf{p_5}$	A_z	CEM	0.250/1	0/1	0.999511	0.628914	0.010184	1.628425	0.989327	0.618730	1.618730	1.618241	61.75511
		H-CEM			0.999756	**0.875686**	0.008367	1.875442	0.991389	0.867319	1.867319	1.867075	104.6595
		E-CEM			0.999756	0.875287	**0.001040**	1.875043	**0.998716**	**0.874247**	**1.874247**	1.874003	**841.6221**
	$A_z^{\text{normalized}}$	CEM			0.999756	0.628914	0.010187	1.628670	0.988886	0.618727	1.618727	1.618486	61.73692
		H-CEM			**1**	**0.875686**	0.008369	**1.875686**	0.991631	0.867317	1.867317	1.867319	104.6345
		E-CEM			**1**	0.875287	0.001041	1.875287	0.998959	0.874246	1.874246	**1.874247**	840.8136

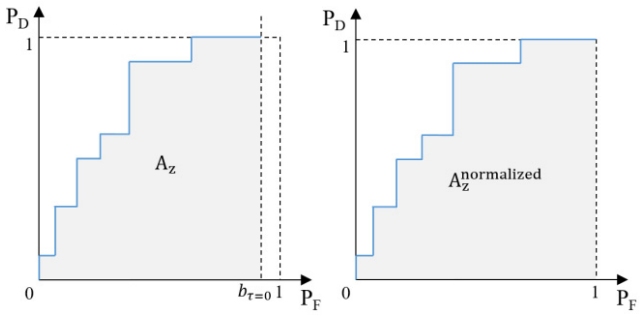


Fig. 13. Original discrete ROC curve and its normalized ROC curve using (9).

[23, Ch. 5], $\text{AUC}_{(F,\tau)}$ is more important than $\text{AUC}_{(D,\tau)}$ in target detection since BKG suppression is more crucial than TD. This fact was also reflected in AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, and AUC_{ODP}. Unfortunately, the BKG suppression issue in hyperspectral target detection has not been discussed in the literature.

As a concluding remark, all experiments performed in this section are particularly designed to demonstrate the full utility of 3D ROC analysis that can be used to evaluate a detector for its effectiveness, TD, BS, TD-BS, ODP, OD, and SNPR from all aspects. In addition, it also illustrates the issues in generating a discrete 2D ROC curve and how to find its correct AUC value by normalization. To do so, the HYDICE data set in Fig. 6 was particularly selected in favor of other data sets in the sense that it is well-studied along with precise knowledge of spatial locations of all 19 R target panel pixels provided by ground truth [19], [23], [29], [32]. For example, as shown in the detection maps for the HYDICE scene in Fig. 7, H-CEM and E-CEM could detected two red panel pixels, p_{211} and p_{221} in row 2 and p_{511} and p_{521} in row 5, lined up vertically and two red panel pixels, p_{311} and p_{312} in row 3 and p_{411} and p_{412} in row 4, lined up horizontally, while CEM could not. In addition, H-CEM and E-CEM could detect all three red subpanel pixels in the third column, but CEM could barely be detected. For such subtle detection, the ground truth provides invaluable information for detectors which performed comparably with their tiny difference not appreciable. Despite the fact that there are also other data sets available in the public domain, many of them are not applicable either with lack of ground truth such as San Diego airport data set or to different applications such as the Purdue data for classification and Houston for data fusion. Nevertheless, the entire analysis conducted in this section is independent of selected data sets and can also be applied to these data sets. Unfortunately, in this case, we may not be able to conduct detailed and comprehensive analysis to appreciate the 3D ROC analysis for what we did for the HYDICE scene in Fig. 6. On the other hand, the target detectors selected for experiments are CEM-based subpixel detectors, CEM [24]–[29], H-CEM [30], and E-CEM [31] for comparison because their performances were shown to be superior to most existing detectors reported in the literature. These experimental results are also new and have never published. Of course, other detectors can also be used for this purpose but will serve for no further value.

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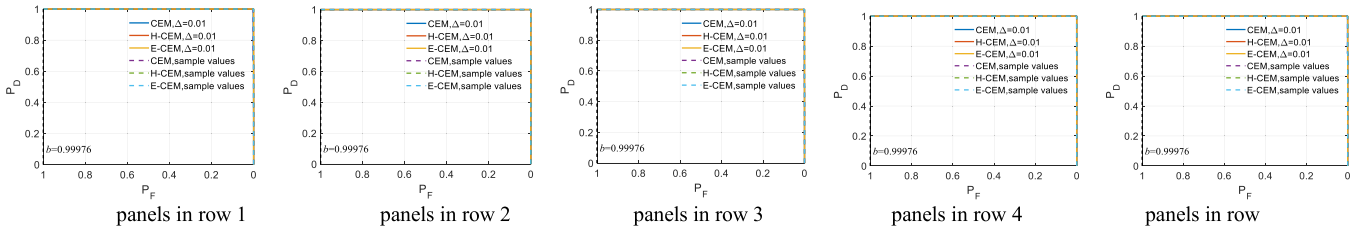
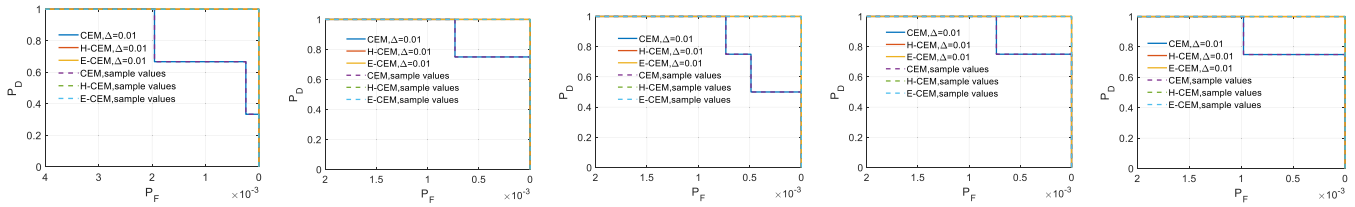


Fig. 14. 2D ROC curves of (P_D, P_F) produced by CEM, H-CEM and E-CEM using (9) by data sample values and uniform step size, $\Delta = 0.01$.



(a) detection maps of CEM

Fig. 15. Normalized 2D ROC curves of (P_D, P_F) produced by CEM, H-CEM, and E-CEM using (9) by data sample values and uniform step size, $\Delta = 0.01$.

TABLE VII

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR ANOMALY DETECTION RESULTS USING (7) AND UNIFORM STEP SIZE, $\Delta = 0.01$

AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
A_z	RX-AD	0/1	0.000125/1	0.98636	0.25071	0.03600	1.23708	0.95036	0.21470	1.21470	1.20107	6.96311
	R-XD			0.98643	0.24833	**0.03586**	1.23477	0.95057	0.21246	1.21246	1.19890	6.92371
	CRD-DW-STO			**0.99654**	**0.43214**	0.05058	1.42868	0.94595	**0.38156**	**1.38156**	1.37810	**8.54324**
$A_z^{normalized}$	RX-AD			0.98649	0.25071	0.03601	1.23720	0.95048	0.21470	1.21470	1.20119	6.96234
	R-XD			0.98656	0.24833	**0.03587**	1.23489	**0.95069**	0.21246	1.21246	1.19902	6.92294
	CRD-DW-STO			**0.99666**	**0.43214**	0.05058	**1.42880**	0.94607	0.38155	**1.38156**	**1.37822**	8.54223

TABLE VIII

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR ANOMALY DETECTION RESULTS USING (7) AND DATA SAMPLE VALUES

AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
A_z	RX-AD	0/1	0.000125/1	0.98724	0.25752	0.03606	1.24476	0.95117	0.22145	1.22145	1.20869	7.14063
	R-XD			0.98717	0.25490	**0.03592**	1.24208	0.95125	0.21898	1.21898	1.20616	7.09646
	CRD-DW-STO			**0.99656**	**0.44686**	0.05060	1.44343	0.94596	**0.39626**	**1.39626**	1.39283	**8.83099**
$A_z^{normalized}$	RX-AD			0.98736	0.25752	0.03606	1.24488	0.95129	0.22145	1.22145	1.20881	7.13964
	R-XD			0.98729	0.25490	**0.03592**	1.24220	**0.95137**	0.21898	1.21898	1.20627	7.09567
	CRD-DW-STO			**0.99669**	**0.44686**	0.05060	**1.44355**	0.94608	0.39625	**1.39626**	**1.39295**	8.82994

VIII. EXPERIMENTS FOR ANOMALY DETECTION

Since anomalies are considered to be very important targets in hyperspectral imaging, anomaly detection has received much interest lately. To evaluate the effectiveness of an anomaly detector, the 2D ROC analysis has also been used as a performance measure. So similar experiments done for prior target detection in Section VII were also performed for anomaly detection. In this case, an image scene which is widely used for anomaly detection was selected for experiments. It is another HYDICE urban image with pseudo-color shown in Fig. 18(a). It is an urban scene and comprised 210 spectral bands with 174 bands being used for experiments after the noise and water absorption bands had been removed. A region with a size of 80 $\times$ 100 pixels located at the upper right of the scene was selected as the test image shown in Fig. 18(b) along with the ground-truth map shown in Fig. 18(c) where 21 pixels were identified as anomalies, which were cars and roofs, because they had spectra that differ from the BKG.

To demonstrate the utility of 3D ROC analysis for anomaly detection, two types of anomaly detectors were used for experiments. One is the classic anomaly detector, RXD developed by Reed and Yu [33], referred to as RX-AD in this article and R-AD in [29] which was originally referred to as R-RXD in [34]. The other is collaborative RXD (CRD) in [35] which has been shown to be superior to many existing anomaly detectors published in the literature where CRD-DW-STO [35] using the size of the outer window, w_{out}, and the size of the inner window, w_{in}, $(w_{\text{out}}, w_{\text{in}}) = (11, 9)$, was shown to be the best selection to be implemented for the HYDICE urban scene [21].

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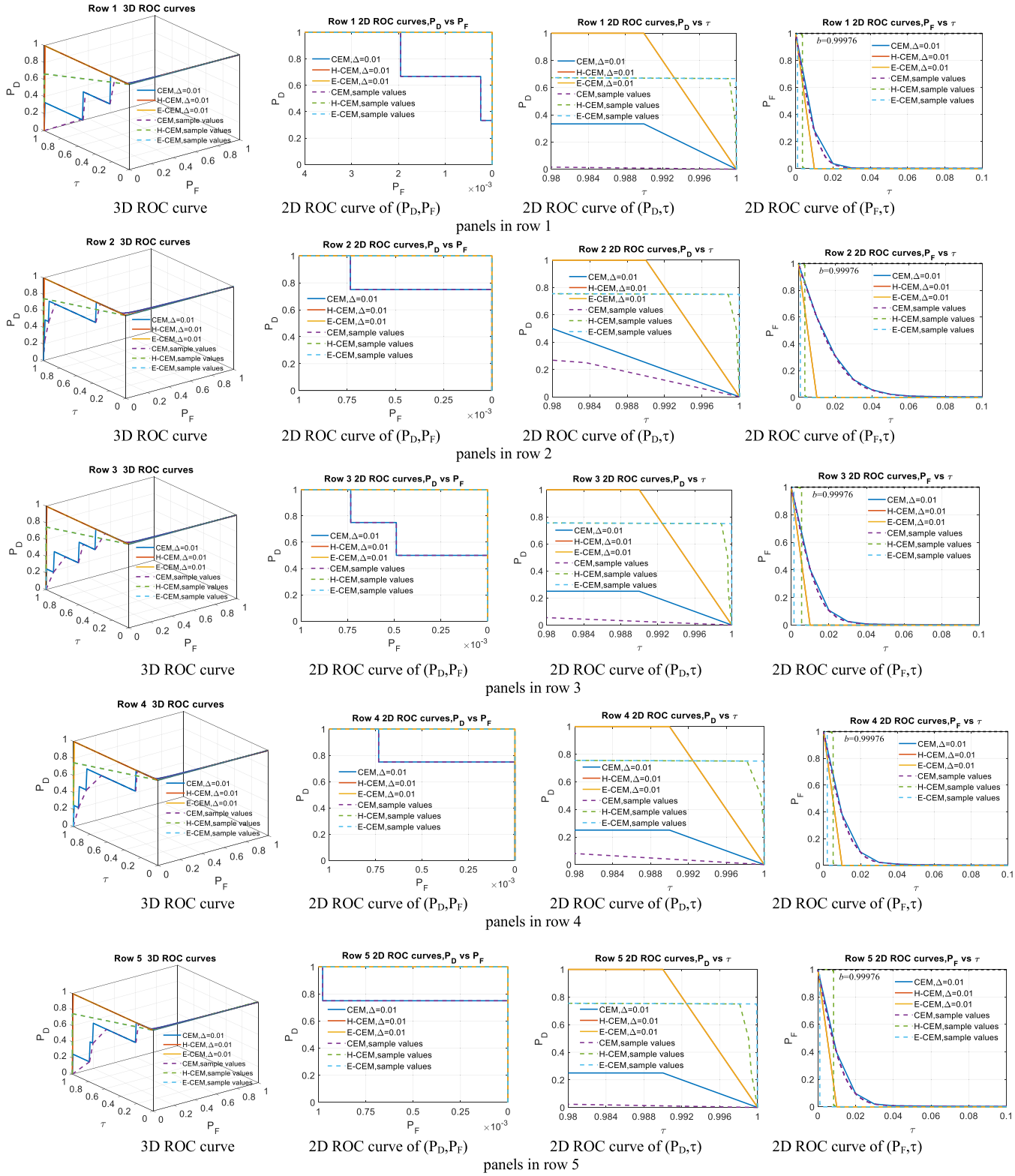


Fig. 16. 3D ROC curves along with their generated three 2D ROC curves using (9) by data sample values and uniform step size, $\Delta = 0.01$.

Fig. 19(a)–(c) shows the anomaly detection maps produced by RX-AD, R-AD, and CRD-DW-STO, respectively.

Apparently, a qualitative analysis based on visual inspection of Fig. 19(a)–(c) did not show much appreciable difference among these three anomaly detectors. In this case, we must rely on a quantitative analysis based on 2D/3D ROC curves produced by the detection maps in Fig. 18 and then further use the eight detection measures developed in Section V-D to calculate their corresponding AUC values, $\mathrm{AUC}_{(\mathrm{D},\tau)}$ $\mathrm{AUC}_{(\mathrm{F},\tau)}$, $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{(\mathrm{F},\tau)}$, and $\mathrm{AUC}_{\mathrm{TD}}$ in (17), $\mathrm{AUC}_{\mathrm{BS}}$ in

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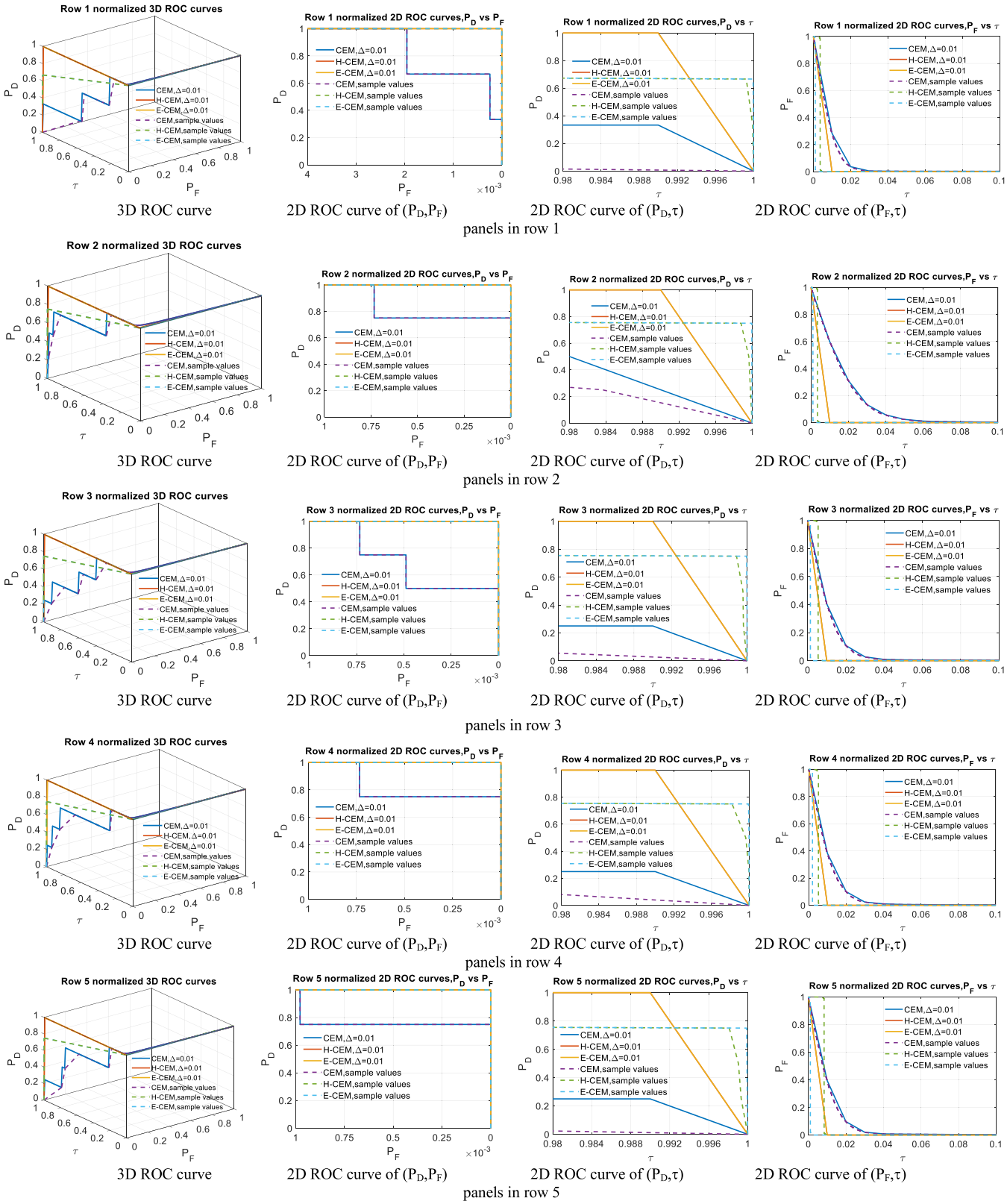


Fig. 17. Normalized 3D ROC curves along with their generated three normalized 2D ROC curves using (9) by data sample values and uniform step size, $\Delta = 0.01$.

(18), $\mathrm{AUC}_{\mathrm{TD-BS}}$ in (19), $\mathrm{AUC}_{\mathrm{ODP}}$ in (20), $\mathrm{AUC}_{\mathrm{SNPR}}$ in (22), and $\mathrm{AUC}_{\mathrm{OD}}$ in (16) to conduct a comparative analysis.

Figs. 20 and 21 show the 3D ROC curves along with their three corresponding 2D curves without and with normalization, respectively, both of which were calculated by (7) from the detection maps produced by RX-AD, R-AD, and CRD-DW-STO using uniform step size $\Delta = 0.01$ and sample values.

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TABLE IX

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR ANOMALY DETECTION RESULTS USING (9) AND UNIFORM STEP SIZE, $\Delta = 0.01$

AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
A_z	RX-AD	0/1	0/0.99987	0.98624	0.25071	0.03600	1.23695	0.95023	0.21470	1.21470	1.2009	6.96331
	R-XD			0.98631	0.24833	**0.03586**	1.23464	0.95044	0.21246	1.21246	1.1987	6.92410
	CRD-DW-STO			**0.99641**	**0.43214**	0.05058	1.42855	0.94583	**0.38156**	**1.38156**	1.37797	**8.54341**
$A_z^{\text{normalized}}$	RX-AD			0.98636	0.25071	0.03600	1.23707	0.95035	0.21470	1.21470	1.20106	6.96253
	R-XD			0.98643	0.24833	**0.03587**	1.23476	**0.95056**	0.21246	1.21246	1.19889	6.92313
	CRD-DW-STO			**0.99654**	**0.43214**	0.05058	**1.42868**	0.94595	0.38155	**1.38156**	**1.37809**	8.54240

TABLE X

AUC VALUES CALCULATED FROM THE THREE 2D ROC CURVES OF (P_D, P_F) FOR ANOMALY DETECTION RESULTS USING (9) AND DATA SAMPLE VALUES

AUCs	Detector	$a_{\tau=1}$ / $a_{\tau=0}$	$b_{\tau=1}$ / $b_{\tau=0}$	$\mathbf{AUC_{(D,F)}}$	$\mathbf{AUC_{(D,\tau)}}$	$\mathbf{AUC_{(F,\tau)}}$	$\mathbf{AUC_{TD}}$	$\mathbf{AUC_{BS}}$	$\mathbf{AUC_{TD-BS}}$	$\mathbf{AUC_{ODP}}$	$\mathbf{AUC_{OD}}$	$\mathbf{AUC_{SNPR}}$
A_z	RX-AD	0/1	0/0.99987	0.98711	0.24655	0.03596	1.23366	0.95114	0.21058	1.21058	1.19769	6.85476
	R-XD			0.98704	0.24425	**0.03582**	1.2313	0.95122	0.20842	1.20842	1.19547	6.81846
	CRD-DW-STO			**0.99644**	**0.41864**	0.05055	**1.41508**	**0.94589**	**0.36809**	**1.36809**	**1.36453**	**8.28165**
$A_z^{\text{normalized}}$	RX-AD			0.98723	0.24655	0.03597	1.23379	0.95126	0.21058	1.21058	1.19781	6.85399
	R-XD			0.98717	0.24425	**0.03582**	1.23142	**0.95134**	0.20842	1.20842	1.19559	6.81751
	CRD-DW-STO			**0.99656**	**0.41864**	0.05055	**1.41521**	0.94601	**0.36808**	**1.36809**	**1.36465**	**8.28067**

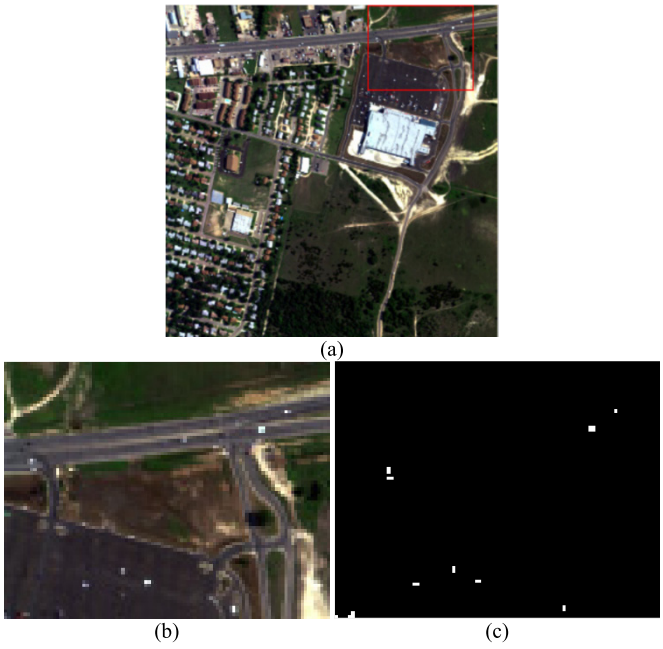


Fig. 18. HYDICE urban scene. (a) Pseudo-color image of the whole scene. (b) Pseudo-color image of the selected area. (c) Ground-truth map.

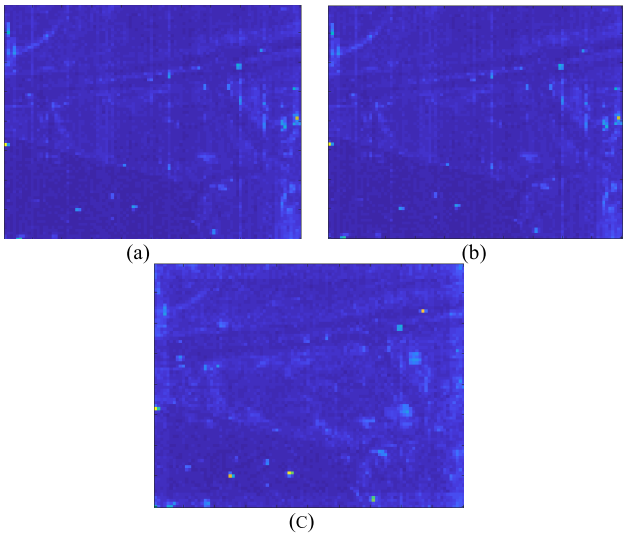


Fig. 19. Detection maps of RX-AD, R-AD, and CRD-DW-STO for HYDICE urban data. (a) Detection map of RX-AD. (b) Detection map of R-AD. (c) Detection map of CRD-DW-STTO.

Similarly, Figs. 22 and 23 show the 3D ROC curves along with their three corresponding 2D curves without and with normalization, respectively, both of which were calculated by (9) from the detection maps produced by RX-AD, R-AD, and CRD-DW-STO using uniform step size $\Delta = 0.01$ and sample values.

Obviously, according to Figs. 19–22, it was impossible to discern 2D ROC curves of (P_D, P_F) among the three anomaly detectors, RX-AD, R-AD, and CRD-DW-STO, without zooming in the range of P_F from 0 to 0.002 since their AUC values were so close. This implied that solely relying on 2D ROC curves of (P_D, P_F) did not have sufficient information to conclude which anomaly detector is better than the others, specifically, RX-AD and R-AD.

However, if we further calculated the values of $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TD-BS}, AUC_{ODP}, AUC_{OD}, and AUC_{SNPD} of Figs. 10–22 and tabulate their results in Tables VII–X with the best results boldfaced, it easily showed from the values of AUC_{TD-BS}, AUC_{OD}, AUC_{SNPD}, and AUC_{ODP} that CRD-DW-STO was a clear winner to outperform the other two, RX-AD and R-AD. Specifically, the value of AUC_{SNPR} provided significant information regarding the detection performance. As for comparison between RX-AD and R-AD,

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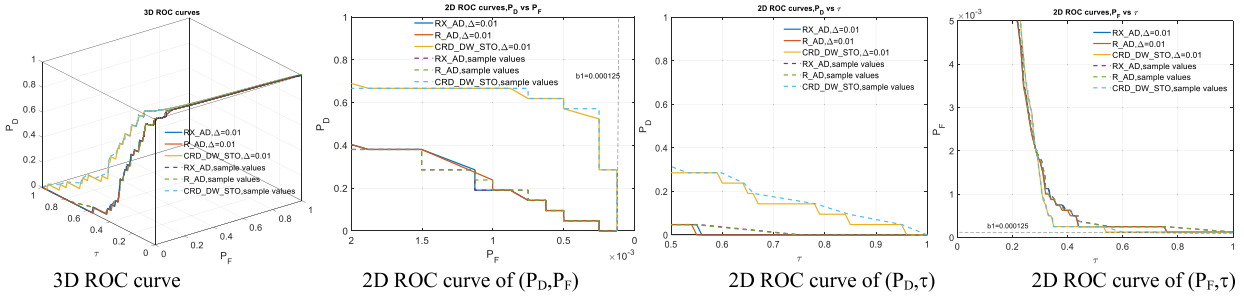


Fig. 20. 3D ROC curves along with their generated three 2D ROC curves produced in Fig. 18 using (7) by data sample values and uniform step size, $\Delta = 0.01$.

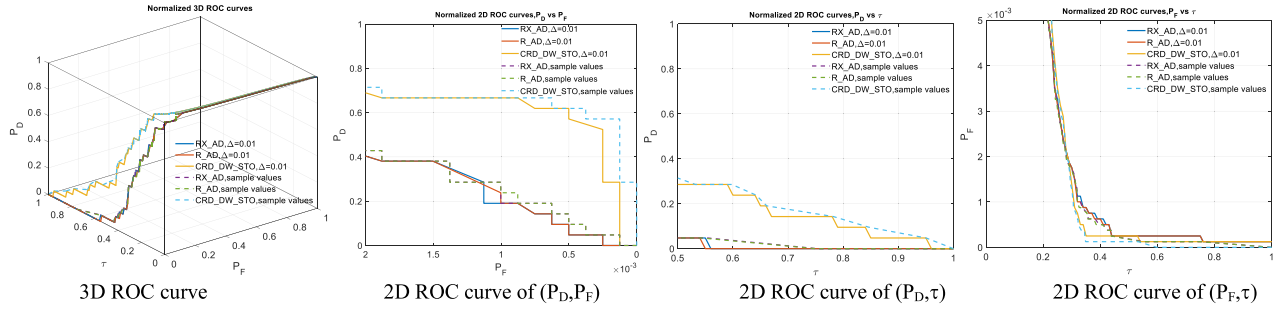


Fig. 21. Normalized 3D ROC curves along with their generated three normalized 2D ROC curves produced in Fig. 18 using (7) by data sample values and uniform step size, $\Delta = 0.01$.

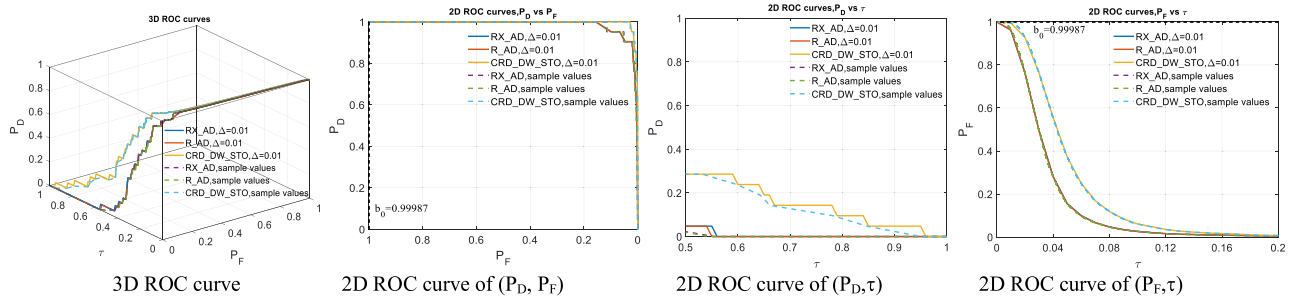


Fig. 22. 3D ROC curves along with their generated three 2D ROC curves produced in Fig. 18 using (9) by data sample values and uniform step size, $\Delta = 0.01$.

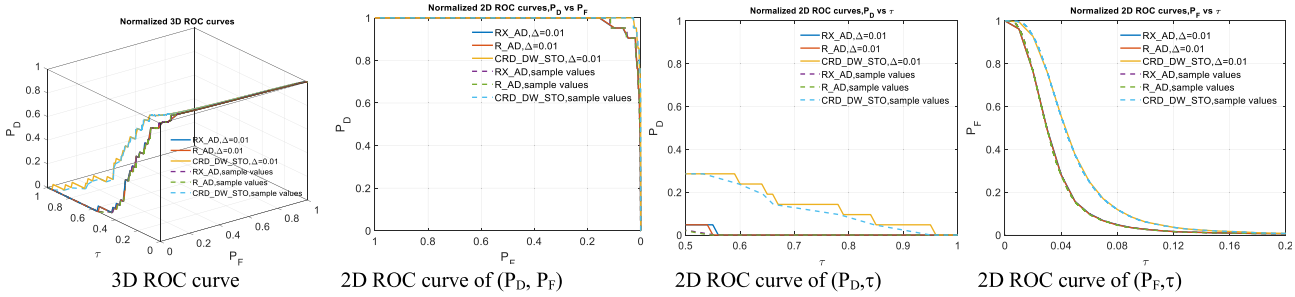


Fig. 23. Normalized 3D ROC curves along with their generated three normalized 2D ROC curves produced in Fig. 18 using (9) by data sample values and uniform step size, $\Delta = 0.01$.

their $\mathrm{AUC}_{(\mathrm{D,F})}$ values were very close within 10^{-4}. In this case, using AUC values of their 2D ROC curves for performance evaluation would not work. However, other detection measures performed more effective than $\mathrm{AUC}_{(\mathrm{D,F})}$. In particular, the $\mathrm{AUC}_{\mathrm{SNPR}}$ values of RX-AD were higher than that of R-AD by nearly 4%.

IX. Conclusion

The ROC analysis has been widely used as an evaluation tool for target and anomaly detection in hyperspectral imaging. Unfortunately, generating discrete ROC curves is not trivial as it is the case for a continuous ROC curve because the data samples used to generate $(P_\mathrm{D}, P_\mathrm{F})$ points on the curve

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are very limited. As a consequence, two issues arise. One is that an ROC curve may look like a step curve rather than a convex-cap curve. In this case, intrapolation and extrapolation are required to complete the curve. However, a complete discrete ROC curve generated by the commonly used linear interpolation is incorrect since the ROC curve is not linear and must be interpolated by a convex-cap curve. This is why all ROC curves reported in the literature are actually zig-zag step-like curves. The other issue is that a discrete 2D ROC curve may not start at $(P_D,P_F) = (0,0)$ and end at $(P_D,P_F) = (1,1)$ as it should be like a continuous ROC curve. Instead, it indeed starts at $(P_D,P_F) = (a_{\tau=1},b_{\tau=1})$ and ends up with $(P_D,P_F) = (a_{\tau=0},b_{\tau=0})$. In this case, normalization should also be performed to obtain the correct AUC value. Interestingly, these two issues are very important but have been overlooked in hyperspectral target/anomaly detection over the past years. Accordingly, one significant contribution of this article is to provide theoretical derivations and practical means of how to generate a discrete ROC curve of (P_D,P_F). On the other hand, an ROC curve of (P_D,P_F) is not sufficient to be used to evaluate detection performance since their results are generated by the same threshold τ as clearly demonstrated in the conducted experiments. In addition, due to the fact that P_D and P_F interact with each other, a high AUC value of an ROC curve of (P_D,P_F) does not necessarily imply a high detection probability or low BS of a detector. To address this dilemma, another significant contribution of this article is to introduce a 3D ROC curve from which three 2D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) can be generated to evaluate the effectiveness, TD, and BS of a detector.

Specifically, six new quantitative detection measures are particularly designed and derived from the AUC values of these three 2D ROC curves. More specifically, $AUC_{(D,F)}$ is developed to evaluate the effectiveness of a detector. $AUC_{(D,\tau)}$ is used to evaluate the TD of a detector, while $AUC_{(F,\tau)}$ is derived to evaluate the BS of a detector. In addition, AUC_{JTDBS} is further deigned to evaluate the TD in BS of a detector with three different versions, AUC_{TD-BS}, AUC_{ODP}, and AUC_{SNPR}. Finally, AUC_{OD} combines all the AUC values to evaluate OD performance of a detector. Except $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$ which have been widely used for target detection [8]–[22], the other six detection measures, AUC_{TD}, AUC_{BS}, AUC_{TD-BS}, AUC_{ODP}, AUC_{OD}, and AUC_{SNPR}, are new and have never been explored in the literature. All these quantitative measures can be jointly used to provide all-round detection performance of a detector. Most interestingly, the idea of SNPR using AUC_{SNPR} follows a similar concept, SNR which is widely used in communications/signal processing. The experimental results showed that SNPR was the most effective detection measure among all the proposed detection measures.

As for future potential research directions, the work presented in this article including the new designed detection measures (AUC_{TD}, AUC_{BS}, AUC_{TD-BS}, AUC_{OD}, AUC_{MDP}, AUC_{OD}, and AUC_{SNPR}) can be extended and applied to many different applications in environmental monitoring, chemical remote sensing, medical data analysis, and medical imaging.

ACKNOWLEDGMENT

The author would like to thank Ms. Shuhan Chen from Zhejiang University, Hangzhou, China, and Mr. Kenneth Yeonkong Ma for their help in performing experiments conducted in this article. The author would also like to specifically thank Shuhan for her effort in preparing 2D/3D ROC software codes uploaded at the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, https://wiki.umbc.edu/display/rssipl/10.+Download.

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